

# Conversation Transcript: User-Identity Induced Political Viewpoints in LLMs

*A conversation between Aryan Ahuja and Claude — August 26, 2026.*

*User and Claude messages are reproduced verbatim. Bracketed italic notes summarize tool activity (web searches/fetches) and stand in for attached documents and images, whose full contents are not reproduced here.*

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## User

tell me if this idea is feasible/how exactly it would work - also feel free to look up current literature via arxiv, openreview, google schol, acl anth, neurips etc. --

Particularly curious about possible political party feature extraction - sae based or more simplistically probing - maybe looking at activations to identify if different steering/intersections result in different activation directions corresponding to different behaviors

## Claude

*[Claude ran web searches on SAE political features, linear representations of political perspective, and activation steering for political bias.]*

Short answer: yes, this is feasible, and it's actually a fairly active corner of the interpretability literature right now — enough that the interesting contribution would be in the specific angle (your "intersections" idea), not in demonstrating that political directions exist at all. Here's the landscape and how you'd actually build it.

### WHAT'S ALREADY ESTABLISHED

The strongest anchor is Kim, Evans & Schein (ICLR 2025). They showed that LLMs possess linear representations of political perspectives within activation space, wherein more similar perspectives are represented closer together, by probing attention heads across the layers of Llama-2-7b-chat, Mistral-7b-instruct, and Vicuna-7b. Key details that matter for your design: non-linear probes did not substantially improve over linear ones, the representations localized in middle layers (~10-20), and probes trained on lawmakers' DW-NOMINATE scores transferred to predicting news-outlet slant with Spearman correlations of 0.72-0.80. That transfer result is important — it means the direction they found is a genuine ideology abstraction, not a memorized lookup of politician names. They also used the probes to identify, monitor, and steer the perspective underlying generated text, so the probe-to-steering pipeline you're sketching is already validated end-to-end at least once.

On the SAE side, there's a very recent paper ("Beyond the Surface: Probing the Ideological Depth of LLMs," Aug 2025) that does almost exactly the SAE half of your idea. They first test steerability (finding some models switch easily between liberal/conservative viewpoints while others resist or refuse), then use SAEs to find latent political features, finding that low-steerability models have more distinct/abstract ideological features — one model had 7.3x more political features than another of similar size — and that ablating a core political feature in an ideologically "deep" model produces consistent, logical shifts in reasoning across related topics, while the same intervention in a "shallow" model just increases refusals. Notably, they frame the SAE work as an advance over probing precisely because SAEs yield a more granular set of features whose causal role can be tested, versus prior work identifying political axes with linear probes.

And on the steering-vector side specifically, there's a small cluster of 2025 work: one paper builds a pipeline using Political Compass Test contrastive pairs to extract layer-wise activations from Mistral and DeepSeek, showing systematic representational bias across layers that can be exploited for steering-vector-based debiasing, with the mechanics being the standard recipe — extract hidden states for contrastive ideological prompt pairs, compute the difference of class means at a target layer (e.g., layer 20), and inject that vector via a forward hook at inference. Follow-ups extend this multilingually with cross-lingual alignment of ideological axes before steering, and separate concurrent work uses a center-of-mass difference of mean attention-head activations for opposing classes and shows it shifts ideological behavior across languages.

#### HOW YOU'D ACTUALLY BUILD IT

Step 1 — Data. You need text/prompts with party or ideology labels. Options: congressional speeches or tweets labeled by party (DW-NOMINATE gives you a continuous score, which is better than binary), Political Compass Test statements as contrastive pairs, or persona prompts ("Answer as a Democrat/Republican would..."). Continuous labels let you train regression probes and check whether the geometry is a scale (moderates in the middle) rather than two clusters — that distinction is itself a finding.

Step 2 — Probing (the cheap baseline, do it first). Run the corpus through an open model (Gemma-2-9B is the pragmatic choice because Gemma Scope gives you pretrained SAEs at every layer for free; Llama-3 also works), cache residual-stream activations at the last token per layer, and fit logistic regression / ridge probes per layer. You'll get a layer-wise accuracy curve; expect the peak in middle layers. Difference-in-means between party classes gives you a steering vector essentially for free from the same cache.

Step 3 — SAE feature extraction. Take the layers where probing peaked, run activations through the SAE, and find features that (a) fire differentially across party-labeled text, or (b) have high cosine similarity between their decoder direction and your probe direction. Then autointerp them (look at max-activating examples via Neuronpedia or your own pipeline). The interesting question is what decomposes: does "Republican" resolve into separate features for, say, religious language, economic policy stances, in-group markers, and rhetorical style? A single probe direction can't tell you that; an SAE can. This is where your idea adds value over the probing literature.

Step 4 — Causal validation (the part that makes it a real result). Correlational feature-firing is weak evidence. You want interventions: clamp/ablate individual SAE features or add the probe direction with varying coefficients, then measure behavior — stance classification of generated text, PCT answers, refusal rate. The ideological-depth paper's ablation-to-"consistent logical shifts across related topics" result is the template.

Step 5 — Your "intersections" question. This is the most novel piece and there are a few concrete versions of it:

- Geometry: compute the angle between the party direction and other directions (religiosity, authoritarianism, economic left-right, sycophancy, refusal). Is "party" a composite of more primitive directions? Can you reconstruct the party probe direction as a sparse linear combination of SAE decoder vectors?
- Compositional steering: steer along direction A and direction B simultaneously and check whether effects compose additively or interfere. E.g., does "economic-left + socially-conservative" steering produce coherent populist outputs, or does the model collapse to one pole? Non-additivity would be evidence the representations aren't independent.
- Steering vs. prompting comparison: does persona-prompting ("you are a Republican") move

activations along the same direction as your extracted vector? Kim et al.'s framework lets you monitor this token-by-token.

## HONEST CAVEATS

Party vs. ideology entanglement is the main conceptual trap — a "Republican" probe trained on 2020s text may really be a Trump-era-rhetoric detector, so test transfer across eras and domains (the DW-NOMINATE-to-media-slant transfer test is the model here). Steering vectors also degrade fluency at high coefficients and effects can be layer- and prompt-sensitive, so report dose-response curves, not single operating points. And ablation results need controls (ablating random features of matched activation frequency) since almost any large intervention changes behavior somehow.

Bottom line: totally feasible with one open model, cached activations, ~one GPU, and existing tooling (TransformerLens/nnsight + SAELens + Gemma Scope). The pure existence claims are published; the open territory is the decomposition/intersection question — whether party identity is one direction, a bundle of separable SAE features, and whether steering along the components behaves compositionally. If you want, I can sketch the actual experiment code skeleton or dig deeper into any of these papers.

## User

could we also use some part of this dataset here:  
<https://huggingface.co/datasets/sarahcen/llm-election-data-2024>

## Claude

*[Claude fetched the Hugging Face dataset page and looked up the associated paper, arXiv:2509.18446.]*

Yes — and it's actually a better fit than it might look at first glance, as long as you're clear about what it is. This is the release from Cen et al.'s longitudinal election study: a large-scale study of 12 models queried with a structured survey of over 12,000 questions on a near-daily cadence from July through November 2024, with systematically varied content and format enabling analyses of behavior over time, sensitivity to steering, responsiveness to instructions, and election-related knowledge and "beliefs," including analyses of longitudinal variation during election season, sensitivity of election-related responses to demographic steering, models' beliefs about candidate attributes, and implicit predictions of the election outcome.

The key thing to understand: this is a dataset of model outputs, not human-labeled partisan text. Each row is (model, date, question, steering/variation metadata, response). And note from the preview that a chunk of it is deliberately non-political control content — the rows I pulled are literally GSM8K math problems tagged category: GSM8K\_multiple\_choice (baseline drift-detection probes), so you'd filter on category/type to get to the election content. Also, the responses come from mostly closed models (claude-3-5-sonnet etc.), so you can't get activations from the models that produced them. That constrains how you use it, but there are three genuinely good uses:

1. The prompt bank is the best part. Their 12k-question survey with systematic variation is exactly the stimulus set you'd otherwise have to build by hand. Crucially, the schema (base\_q\_template, variation\_type, prompt\_type, pre\_prompt\_q) gives you matched pairs: the same base question under different steering conditions (including demographic/partisan persona steering). That's the ideal input for difference-in-means steering vectors and probes, because content is held constant and only the persona/framing variable changes — so your extracted direction isn't confounded by

topic. Run these prompt variants through your open model (Gemma-2 + Gemma Scope), cache activations, and compute per-condition directions. This directly enables your "intersections" question: you get separate directions for party persona, demographic persona, and question framing from the same matched design, and can measure angles between them and test compositional steering.

2. Responses as labeled text for probe validation. Use the steered responses (e.g., outputs generated under Democrat-leaning vs Republican-leaning steering) as an out-of-distribution test set: does your probe/SAE feature, trained on congressional text or PCT pairs, correctly separate them when you run the responses through your open model? That's a strong generalization check, and it connects prompt-level steering (their manipulation) to activation-space geometry (your measurement) — i.e., does persona prompting move text along the same internal direction as your extracted vector?

3. Behavioral steerability as external ground truth. Their paper quantifies per-model sensitivity to steering behaviorally. The ideological-depth paper I mentioned earlier claims steerability anti-correlates with the distinctness of internal political features. If any of their 12 models are open-weight (worth checking the model list — if Llama or similar is in there, you're in luck), you could directly test that behavioral-steerability / internal-feature-structure link using their longitudinal behavioral data on one side and your SAE analysis on the other. Even for closed models, their steerability rankings give you a behavioral benchmark to compare against your open-model internal findings.

Two caveats. First, RLHF'd chat-model responses are heavily hedged and refusal-laden, so an "ideology" direction extracted from responses alone risks being contaminated by refusal/hedging directions — another reason the prompts are the cleaner asset, and also a reason to explicitly measure the angle between your party direction and a refusal direction (which is itself an interesting intersection result). Second, everything is frozen in the 2024 election context, so it's great as a test/validation domain but you'd want your training corpus for probes to be broader (congressional speech across eras) to avoid learning a Trump-vs-Harris-2024 detector rather than a party representation.

So the practical pipeline: filter out baseline categories, group rows by `base_q_template_hash` to reconstruct matched steering pairs, feed prompt variants through Gemma-2, extract directions/SAE features from the matched contrast, validate on their response text and your external corpora. Want me to write the data-wrangling code to reconstruct those matched pairs from the HF dataset?

## User

what would we need to do for a proof of concept for this - could we run the os models and analysis locally

## Claude

Yes, comfortably — this is one of the cheaper interp projects you could pick, because the expensive part (training SAEs) is already done for you. Here's the realistic breakdown.

### HARDWARE REALITY CHECK

The trick is to use a small model with pretrained public SAEs. The canonical choice is Gemma-2-2B + Gemma Scope (DeepMind's released SAE suite covering every layer). Rough numbers:

- Gemma-2-2B in bf16: ~5 GB VRAM. Runs on any consumer GPU (RTX 3060+), Apple Silicon with 16 GB unified memory, or free Colab T4. Add one SAE (~few hundred MB per layer) and you're still under 8 GB.
- Gemma-2-9B if you want a more capable model: ~18-20 GB bf16, so a 24 GB card (3090/4090) or 4-bit quantized on 12 GB. Gemma Scope covers 9B too.
- Probing is trivial: once activations are cached, it's sklearn logistic regression on your CPU. Seconds.
- Activation caching for a PoC corpus (say 2-5k prompts, one residual stream vector per prompt at the last token, across ~26 layers of the 2B model) is a few hundred MB on disk. Nothing.

A MacBook with 16 GB can do the whole PoC; a single 3090/4090 makes it pleasant. No cluster needed. Steering (generation with a hooked forward pass) is the slowest part and even that is just normal inference speed.

## TOOLING

`transformer_lens` (or `nnsight`) for hooked activation access, `sae_lens` for loading Gemma Scope SAEs in two lines, datasets for pulling the HF election data, sklearn for probes. Neuronpedia is also worth knowing about: Gemma Scope features are all browsable there with auto-interpretations, so you can look up what a feature "means" without running any autointerp pipeline yourself — for a PoC that saves you an entire subsystem.

MINIMAL POC (roughly a weekend of work, in order)

Step 1 — Data (~2-4 hrs). Pull the Cen et al. dataset, filter out the GSM8K/baseline categories, and group by `base_q_template_hash` to reconstruct matched pairs: same question, different partisan/demographic steering condition. Target maybe 500-1,000 matched pairs. Optionally supplement with a small set of clearly-labeled partisan text (e.g., a few hundred congressional statements per party) as a second, independent corpus — you want two corpora so you can test whether the direction transfers.

Step 2 — Cache activations (~1 hr of compute). Run every prompt variant through Gemma-2-2B, save the residual stream at the final token for all layers. One forward pass per prompt, no generation.

Step 3 — Probe + direction (~1 hr). Train a logistic probe per layer on party condition; plot accuracy vs. layer (expect a mid-layer peak — this plot alone is your "does the model represent this?" sanity check). Compute the difference-in-means vector at the best layer. Check that the probe trained on the election prompts transfers to the congressional corpus and vice versa — if it doesn't, you've learned a 2024-election detector, not a party direction, and you know it immediately.

Step 4 — SAE lens (~2-3 hrs). Push the same activations through the Gemma Scope SAE at your peak layer. Two views: (a) which features fire differentially across party conditions (simple t-test / mean-difference across features), (b) which features' decoder vectors have high cosine similarity with your probe direction. Take the top ~20, look them up on Neuronpedia. This is where the fun result lives: does the party direction decompose into interpretable sub-features (religion, economics, group identity, rhetorical style) or is it one monolithic feature?

Step 5 — Causal check (~2-3 hrs). Add the difference-in-means vector (and separately, clamp the top SAE features) during generation on neutral political prompts, sweep the coefficient, and classify the outputs' slant (a zero-shot stance classifier or even GPT/Claude-as-judge on a small sample works for a PoC). A dose-response curve — output slant shifting monotonically with

steering coefficient until fluency breaks — is the "it's causal, not just correlational" evidence. As a control, steer along a random vector of matched norm and show it doesn't shift slant.

Step 6 — The intersection teaser (optional but the novel bit). Extract one more direction from the same cache — refusal (refuse/comply contrast pairs are easy to generate) or a demographic direction from the dataset's own steering conditions — and report the cosine angle with the party direction, plus what happens when you steer along both simultaneously. Even a single "party vs refusal? party vs demographic-persona?" measurement makes the PoC say something new rather than replicating Kim et al.

#### WHAT YOU'RE NOT DOING IN THE POC

No SAE training, no autointerp pipeline, no multi-model comparison, no large-scale generation evals. Those are all phase-2 items once the layer-wise probe curve and one steering dose-response plot confirm the machinery works.

Total compute for the whole thing is maybe 2-4 GPU-hours on a 3090-class card. The scarce resource is your time on data cleaning and eval judgment calls, not FLOPs.

If you want, I can write the actual scaffold — the pair-reconstruction code for the HF dataset plus the TransformerLens caching/probing script — so you have something runnable to start from.

### User

i've started a write-up - how does this look: This doc will serve as documentation for an attempt at proof-of-concept for an idea I had during Professor Chara Podimata's talk at Penn's ASSET Seminar. This talk centered around her longitudinal study of LLMs in response to election-centered questions, studying political bias and LLM priors in response to some simulated user to pinpoint the way in which chatbots may affect information dissemination among potential voters. This proof-of-concept attempts to determine whether context given about a user meaningfully shifts the reasoning pathways a model takes to answer a political question. It also provides a proxy for whether this shift, if it exists, may be construed as some high-level differentiation in how LLMs may conceptualize different users: or rather, why Professor Podimata's research group's results show the findings that they do.

#### The Idea

##### Priors

Professor Podimata's study of blackbox frontier models and their responses to simulated potential voters found particularly interesting, and sometimes concerning, results that seemed to stem from the context given about the simulated voter. The research group referred to this contextual stimulus as "steering". (For the purpose of preserving the word "steering" to instead signify alteration of a model's behavior via its internals, as is the case in standard mechanistic interpretability literature, I'll refer to this context as "user scaffolding" instead). This user scaffolding often swayed the model's response regarding political topics, such as describing Donald Trump as a "family man" to a user with the scaffolded identity of a woman as opposed to describing Donald Trump as a "patriot" to a user with the scaffolded identity of a man. In the interest of brevity, I'll stop here, though the most important note to take is that differing user context invokes heavily differing behaviors in the model, ostensibly centering around the user's supposed societal norms (e.g. women are family-oriented and men are gung-ho).

Additionally, existing literature has found that political viewpoints are represented linearly as a high-level concept in model activations and are localized to middle layers much like the quintessential concept of refusal. Further, viewpoints that are similar (say, belonging to the same

political party), tend to cluster together. Vice versa for viewpoints that are ideologically distinct. Other work has followed inter-layer political bias within models and political feature extraction a la SAEs; correspondingly the research will mess with internals to debias via steering or ablate certain political features to shift the model's ideological positioning.

Now

## Claude

This is a solid start — the framing (talk, observation, mechanistic question) is a genuinely good motivation structure, and the "user scaffolding" terminology decision is smart and worth keeping. Feedback, roughly ordered by importance:

1. Sharpen the actual hypothesis. Right now the core claim is "whether context given about a user meaningfully shifts the reasoning pathways a model takes." Two issues: "reasoning pathways" overpromises — what you're actually measuring in the PoC is representational shift (do persona-scaffolded prompts move activations along identifiable directions, e.g., an ideology direction, and do different scaffolds map to distinct directions?). Circuit-level "pathways" would be a different, harder project. Second, the sentence "It also provides a proxy for whether this shift... may be construed as some high-level differentiation in how LLMs may conceptualize different users" is doing a lot of hedging ("proxy," "may be construed," "some," "may"). I'd replace both with something like a crisp two-part hypothesis: (H1) user scaffolding produces measurable, systematic displacement in activation space along interpretable directions; (H2) the direction of displacement predicts the behavioral differences Podimata et al. observed. That gives every later experiment something to be evidence for or against.

2. Be careful with the empirical example. The "family man" vs. "patriot" example is vivid, but if it's your recollection from the talk rather than a result you can point to in the paper, flag it as such ("in one example presented, ..."), or pull the actual candidate-attribute findings from the paper (arXiv:2509.18446 — their analysis (iii) on candidate attributes and (ii) on demographic steering sensitivity is where this lives). In a doc that will be read by others, one misremembered example undermines the carefully-hedged stuff around it. Same paragraph: "ostensibly centering around the user's supposed societal norms (e.g. women are family-oriented and men are gung-ho)" — the parenthetical reads as you endorsing the stereotype framing rather than attributing it to the model. Something like "(i.e., the model appears to condition its framing on demographic stereotypes)" keeps the observation and drops the ambiguity. "Gung-ho" also probably isn't the register you want.

3. Tighten the priors paragraph and add citations. "Existing literature has found that political viewpoints are represented linearly... localized to middle layers much like the quintessential concept of refusal" — good, but name it: Kim, Evans & Schein (ICLR 2025) for the linear representations (worth noting they probed attention heads, layers ~10-20), Arditi et al. 2024 for the refusal direction. "Vice versa for viewpoints that are ideologically distinct" is logically dangling — the first clause says similar viewpoints cluster, so the vice versa should be "distant viewpoints are represented farther apart," but as written it's ambiguous. Just say it explicitly. And "the research will mess with internals to debias via steering" — "mess with" is too casual for a doc, and this sentence conflates two separate literatures (steering-vector debiasing, e.g. the PCT contrastive-pair work; and SAE feature ablation, e.g. the ideological-depth paper). One sentence each, with cites, sets up your contribution cleanly.

4. Add the "gap" sentence before "Now." Classic structure: after summarizing priors, one sentence stating what's missing that you're doing — e.g., prior work extracts ideology directions from

content (partisan text, PCT statements), but no one has tested whether user-identity scaffolding moves activations along those same directions, which is precisely the mechanism that would explain the longitudinal study's behavioral findings. That's your bridge, and it's actually a real gap, so claim it explicitly.

5. Small line edits. "attempt at proof-of-concept" -> "a proof-of-concept attempt"; "particularly interesting, and sometimes concerning, results that seemed to stem from" — cut one adjective clause, it reads smoother; "In the interest of brevity, I'll stop here, though the most important note to take is" -> "The key observation:"; "invokes heavily differing behaviors" -> "elicits substantially different behavior." Also decide on person/register — you're currently mixing academic distance ("this proof-of-concept attempts to determine") with blog voice ("I'll stop here," "mess with"); either is fine for a working doc, but pick one.

One structural suggestion for whatever comes after "Now": state the concrete PoC design (model, SAE suite, dataset, the matched-pair reconstruction from the Cen et al. HF release) before any results, and register your predictions in advance — including what a null result looks like (scaffolding shifts activations but along directions orthogonal to ideology, which would itself be an interesting finding about persona vs. ideology being decoupled internally). Pre-stating the null makes the eventual write-up credible either way.

Want me to do a tracked-changes-style rewrite of the two sections you've got, or leave it at feedback?

## User

how is this so far: Aryan Ahuja  
Penn Engineering  
8/26/26

This doc will serve as documentation for an attempt at proof-of-concept for an idea I had during Professor Chara Podimata's talk at Penn's ASSET Seminar. This talk centered around her longitudinal study of LLMs in response to election-centered questions, studying political bias and LLM priors in response to some simulated user to pinpoint the way in which chatbots may affect information dissemination among potential voters ([Cen et al. 2025](<https://arxiv.org/pdf/2509.18446v1>)). This proof-of-concept attempts to determine whether differing the context given about a user meaningfully shifts model activations along identifiable directions (say an ideology direction) or if different scaffolds map to different directions. I attempt to find whether it is worth exploring why Professor Podimata's research group's results show the findings that they do.

### The Idea

#### Priors

Professor Podimata's study of blackbox frontier models and their responses to simulated potential voters found particularly interesting, and sometimes concerning, results that seemed to stem from the context given about the simulated voter. The research group referred to this contextual stimulus as "steering". (For the purpose of preserving the word "steering" to instead signify alteration of a model's behavior via its internals, as is the case in standard mechanistic interpretability literature, I'll refer to this context as "user scaffolding" instead). This user scaffolding often swayed the model's response regarding political topics. In an example presented, the model described Donald Trump as a "family man" to a user with the scaffolded identity of a woman as opposed to describing Donald Trump as a "patriot" to a user with the scaffolded identity of a man. In the interest of brevity, I'll stop here, though the most important



note to take is that differing user context invokes heavily differing behaviors in the model (i.e., at least in this case, the model appears to condition its framing on demographic stereotypes).

Additionally, existing literature has found that political viewpoints are represented linearly as a high-level concept in model activations and are localized to middle layers much like the quintessential concept of refusal. Further, viewpoints that are similar (say, belonging to the same political party), tend to cluster together. Vice versa for distant viewpoints, which are represented farther apart. Other work has

1. Followed inter-layer political bias within models, and correspondingly utilized steering-vectors to debias the models
2. Extracted latent political features a la SAE, followed by ablating a core political feature to shift the model's ideological positioning.

Now

This p-o-c has two hypotheses:

1. User scaffolding leads to measurable and systematic displacement of activations along interpretable directions
2. This direction of displacement predicts the behavioral differences found by Podimata et. al.

Methodology

## Claude

Much improved — the hypotheses are now crisp and falsifiable, the stereotype observation is properly attributed to the model, and the example is flagged as "an example presented."

Remaining issues, in descending order:

H2 isn't testable as written. "Predicts the behavioral differences found by Podimata et al." — their behavioral data comes from closed frontier models, and you can't get activations from those, so your open-model displacement direction can't literally predict their measurements. You need a testable proxy, something like: "H2: the displacement direction from user scaffolding aligns with (has high cosine similarity to) an independently extracted ideology direction, and steering along it reproduces, in an open model, the same kind of framing shifts observed behaviorally in Cen et al." That keeps the spirit — connecting your mechanism to their phenomenon — while being something you can actually measure. Alternatively, split it: H2a (alignment with ideology direction) and H2b (causal steering along that direction shifts output framing in the predicted direction). Worth also pre-registering the interesting null: scaffolding displaces activations along directions orthogonal to ideology, which would suggest persona and ideology are decoupled internally — that's a publishable-flavored finding, not a failure.

Attribution consistency. You cite Cen et al. 2025 in the intro but write "Podimata et al." in H2. Pick one — the paper is Cen et al. (Podimata is co-last author), so "Cen et al." for the paper and "Podimata's talk" for the talk is the clean split you've already half-established. Keep it consistent throughout.

The intro's last sentence is circular. "I attempt to find whether it is worth exploring why... results show the findings that they do" says almost nothing — you're determining whether it's worth exploring an explanation, which is a meta-question. Replace with the actual stake: something like "If such a shift exists and aligns with known ideology representations, it offers a mechanistic explanation for the behavioral sensitivity to user context that Cen et al. document." Direct claim, direct payoff.

Priors still have no citations. "Existing literature has found that political viewpoints are represented linearly..." needs Kim, Evans & Schein (ICLR 2025) — and one nuance worth adding since it affects your methodology: they probed attention-head activations at layers ~10-20, so if you probe the residual stream instead, note that as a deliberate simplification. The refusal comparison needs Ardit et al. 2024. Your numbered items 1 and 2 need their cites too (the PCT steering-vector work, e.g. Nadeem et al. 2025, and the ideological-depth SAE paper respectively). For a doc positioning itself against prior work, uncited priors read as folklore.

Still missing the gap sentence. After the numbered list and before "Now," you want one sentence claiming your lane: prior work extracts political directions from content (partisan text, PCT statements) — nobody has tested whether user-identity scaffolding moves activations along those same directions. That's the sentence that makes hypotheses 1 and 2 feel inevitable rather than arbitrary.

Small stuff: "an attempt at proof-of-concept" -> "a proof-of-concept attempt"; "differing the context" -> "varying the context"; "invokes heavily differing behaviors" -> "elicits substantially different behavior"; "a la" -> "à la"; "p-o-c" vs "proof-of-concept" — pick one (spell it out at first use, "PoC" after); "et. al." -> "et al." (no period after "et").

For the Methodology section you're about to write, the skeleton that follows naturally from your hypotheses: (1) models & tools — Gemma-2-2B/9B, Gemma Scope SAEs, TransformerLens/SAELens; (2) data — matched scaffold pairs reconstructed from the Cen et al. HF release via base\_q\_template\_hash, plus an independent ideology corpus for the reference direction; (3) direction extraction — difference-in-means + linear probes per layer; (4) H1 test — displacement magnitude vs. matched controls; (5) H2 test — cosine alignment with the reference ideology direction + SAE feature decomposition; (6) causal check — steering dose-response with a random-direction control. Each hypothesis gets exactly one measurement that could kill it, which is the property you want a reader to notice.

## User

tell me exactly where i need to cite and find the most updated links that i need to cite

## Claude

*[Claude ran verification searches for the ideological-depth paper's authors/versions and the refusal-direction paper.]*

Here's the sentence-by-sentence citation map for your draft, with verified links (all current as of today):

### Intro paragraph

1. "her longitudinal study of LLMs in response to election-centered questions..." — you already have this one, but two fixes: cite the abstract page rather than the versioned PDF (so the link survives revisions), and note the correct format is Cen et al. 2025, arXiv:2509.18446 -> <https://arxiv.org/abs/2509.18446>. Since the paper's data is what you'll use, also cite the dataset release itself where you first mention using it: <https://huggingface.co/datasets/sarahcen/llm-election-data-2024>

### Priors, paragraph 1

2. "In an example presented, the model described Donald Trump as a 'family man'..." — this needs either (a) a pointer to the specific analysis in Cen et al. (their candidate-attributes analysis,

analysis iii, and demographic-steering analysis, analysis ii, in arXiv:2509.18446), or (b) an explicit "(example as presented in the ASSET Seminar talk, [date])" if you're citing the talk rather than the paper. Talks are citable; unattributed vivid examples are not. I'd verify the example appears in the paper before attributing it there.

Priors, paragraph 2 — this is where you're most exposed. Four claims, four cites:

3. "political viewpoints are represented linearly as a high-level concept in model activations and are localized to middle layers" + "viewpoints that are similar... tend to cluster together" — both claims are Kim, Evans & Schein, ICLR 2025, "Linear Representations of Political Perspective Emerge in Large Language Models" -> <https://arxiv.org/abs/2503.02080> (OpenReview: <https://openreview.net/forum?id=rwqShzb9li>; code: <https://github.com/JunsolKim/RepresentationPoliticalLLM>). One accuracy note: they probed attention heads across the layers — not the residual stream — so if your methodology probes the residual stream, add a clause acknowledging the difference.

4. "much like the quintessential concept of refusal" — Arditi et al., NeurIPS 2024, "Refusal in Language Models Is Mediated by a Single Direction" -> <https://arxiv.org/abs/2406.11717>. It shows refusal is mediated by a one-dimensional subspace across 13 open-source chat models up to 72B, which is exactly the "quintessential linear concept" you're invoking.

5. Numbered item 1: "Followed inter-layer political bias within models, and correspondingly utilized steering-vectors to debias" — Nadeem et al. 2025, "Steering Towards Fairness: Mitigating Political Bias in LLMs" -> <https://arxiv.org/abs/2508.08846> (v3, Sep 2025, is current). Optionally add the concurrent multilingual identification-and-steering paper for breadth: <https://arxiv.org/abs/2507.22623>

6. Numbered item 2: "Extracted latent political features à la SAE, followed by ablating a core political feature" — Kabir, Esterling & Dong 2025, "Beyond the Surface: Probing the Ideological Depth of Large Language Models" -> <https://arxiv.org/abs/2508.21448>. Heads up that v2 (revised 14 Nov 2025) is the current version, and its framing tightened from v1 — v2 defines ideological depth as (i) steerability and (ii) SAE-measured feature richness, compares Llama-3.1-8B-Instruct and Gemma-2-9B-IT, and finds Gemma activates ~7.3x more distinct political features, with causal ablations of a compact political feature set increasing refusal rates — supporting the view that refusals can stem from insufficient representations, not only safety policy. This paper is unusually load-bearing for you (it's the closest prior work AND it uses Gemma-2-9B, your likely model), so make sure your description matches v2, not the v1 abstract.

Methodology section (when you write it) — tool/artifact cites:

7. Gemma Scope (the pretrained SAEs) — Lieberum et al. 2024 -> <https://arxiv.org/abs/2408.05147>

8. Gemma 2 (the model) — Gemma Team 2024 -> <https://arxiv.org/abs/2408.00118>

9. TransformerLens — cite the repo: <https://github.com/TransformerLensOrg/TransformerLens>; SAEs: <https://github.com/jbloomAus/SAELens>

10. Difference-in-means / contrastive activation steering as a method — if you use it, the standard cites are Arditi et al. (already #4) plus Rimsky/Panickssery et al.'s Contrastive Activation Addition -> <https://arxiv.org/abs/2312.06681>

Two general notes. First, always link arXiv abstract pages (/abs/), never /pdf/...vN — versioned PDF links go stale and abstract pages resolve to the latest version. Second, for the ICLR/NeurIPS papers, the arXiv link is fine for a working doc, but if this graduates to anything formal, the venue versions are: Kim et al. -> ICLR 2025 proceedings (OpenReview link above), Arditi et al. -> NeurIPS

2024 proceedings.

The one claim in your draft that has no possible citation and needs rewording instead: "I attempt to find whether it is worth exploring why..." — as discussed, that's your gap statement, and gaps are argued, not cited.

### User

is refusal also on the attention head activations or just the residual stream

### Claude

Refusal, as established in Arditi et al., is a residual stream result. Their method computes difference-in-means between harmful and harmless instruction activations in the residual stream (sweeping over layers and token positions to pick the single best direction), and their interventions operate on the residual stream too — directional ablation projects the direction out of the residual stream at every layer, and their rank-one weight edit orthogonalizes every matrix that writes into the residual stream (attention output and MLP output weights) against that direction. So the claim "refusal is mediated by a single direction" is a claim about residual-stream geometry.

Kim, Evans & Schein, by contrast, probed attention head outputs — following the Inference-Time Intervention style of analysis (Li et al. 2024, arXiv:2306.03341), where you probe each head's output vector separately and find which heads carry the concept. That's why I flagged the mismatch for your write-up.

The important conceptual point, though, is that these aren't competing ontologies — the residual stream at layer  $L$  is literally the sum of the embedding plus every attention head's and MLP's contributions up to  $L$ . A direction "in the residual stream" got written there by specific heads and MLPs; a concept found "in attention heads" necessarily shows up in the residual stream downstream of those heads. So the two framings are different resolutions of the same object: head-level probing tells you which components write the concept, residual-stream probing tells you the aggregate direction that ends up representing it. (There is follow-up work localizing safety/refusal behavior to specific attention heads — e.g., work on ablating individual safety-relevant heads — which is exactly the head-level view of the same phenomenon.)

For your PoC this cashes out as a practical simplification you can state in one sentence: you'll probe the residual stream (coarser, simpler, and what Gemma Scope SAEs are trained on — most public Gemma Scope SAEs target residual stream sites, with some MLP/attention variants), whereas Kim et al. worked at head resolution; since head outputs sum into the residual stream, a linearly decodable concept at head level implies decodability in the residual stream, so your setup is a valid — if lower-resolution — test of the same representational claim. If the PoC works, drilling down to head-level attribution (which heads write your scaffolding-displacement direction?) is a natural phase-2 extension and would connect your result back to their methodology directly.

### User

can you walk me through the methodology and talk through the tradeoffs, why we choose certain models (i.e. having the existing pre-trained SAEs makes it simpler for a PoC) also are there any existing figures in any of the research papers mentioned that i should include? feel free to download/curl the pdfs and look through for figures in any way you want whether its turning the pages into images etc)

## Claude

*[Claude noted the sandbox network allowlist blocks arxiv.org for curl, and instead fetched the full text (including figure captions) of Kim et al. (arXiv:2503.02080) and Cen et al. (arXiv:2509.18446) via web fetch.]*

Quick note first: my sandbox's network allowlist doesn't include arxiv.org, so I couldn't curl the PDFs into the container — but I pulled the full text of the two most important papers (Kim et al. and Cen et al.) through web fetch, including every figure caption, so I can give you a concrete figure inventory. That fetch also turned up methodology details that change some of my earlier advice for the better, so I'll fold those in.

### METHODOLOGY WALKTHROUGH, DECISION BY DECISION

Decision 1: Which model. Gemma-2-2B (or 9B) over Llama/Mistral/etc. Three reasons stack up. First, Gemma Scope: DeepMind released trained SAEs for every layer of both Gemma-2 sizes, which deletes the single most expensive subsystem from your project — training a decent SAE yourself is days of GPU time and a research problem in its own right (choosing width, sparsity penalty, dead-feature handling), all of which is irrelevant to your actual hypotheses. Second, Kabir et al. used Gemma-2-9B-IT as one of their two candidates and found it activates  $\sim 7.3\times$  more distinct political features than Llama-3.1-8B and is more steerable in both directions — meaning Gemma is the model where political features are known to be rich and manipulable, so a null result on Gemma is informative (the features exist; scaffolding just doesn't engage them) rather than ambiguous (maybe the model just has impoverished political representations). Third, Neuronpedia hosts auto-interpretations for Gemma Scope features, so "what does feature 4231 mean" is a lookup, not a pipeline. The tradeoff: 2B is fast and fits anywhere but is a weaker instruction-follower, which matters because your prompts are chat-style user-scaffolded queries; 9B behaves more like the frontier models Cen et al. studied but needs  $\sim 20$  GB or quantization. Reasonable plan: develop on 2B, confirm headline results on 9B.

Decision 2: Probe target — residual stream vs. attention heads. Kim et al. probed individual attention heads (1,024 separate probes for a  $32\times 32$  model) because they wanted component-level localization and were following the ITI methodology. You should probe the residual stream instead. Tradeoffs: you lose the ability to say which heads write the ideology direction, but you gain (a) 26 probes instead of  $\sim 1,000$ , (b) direct compatibility with Gemma Scope, whose main SAE suite targets residual stream sites, and (c) the mathematically sufficient test — head outputs sum into the residual stream, so anything linearly decodable at head level is present there. State this as a deliberate simplification in the write-up; head-level attribution is your phase-2.

Decision 3: Direction extraction — difference-in-means vs. probe weights vs. SAE features. Do all three, cheapest first, because they answer different questions and cross-validate each other. Difference-in-means (mean activation under "I am a Democrat" scaffolds minus mean under "I am a Republican" scaffolds, per layer) is a one-liner off the cached activations and is the method behind both the refusal result and the PCT debiasing work. Logistic/ridge probes give you an accuracy-vs-layer curve — your "does the representation exist and where" figure — and probe weights double as a second direction estimate; if difference-in-means and probe direction have high cosine similarity, that's a robustness point for free. SAE features are the decomposition step: project the displacement vector onto Gemma Scope's decoder directions and see whether scaffolding moves activations along a few interpretable features or diffusely. Tradeoff to be honest about in the doc: difference-in-means is noisy with small  $n$  and sensitive to confounds in the contrast set (which the matched-pair structure of the Cen data largely handles); probes can latch onto surface features (token identity of "Democrat" vs "Republican" — see Decision 5); SAE

decompositions inherit whatever the SAE failed to learn.

Decision 4: Data. Reconstruct matched pairs from the HF release via `base_q_template_hash`. Something I confirmed from the paper that strengthens this: their prompt variations are clean prefixes like "I am a Democrat" / "I am [ethnicity]" prepended to an otherwise identical question — 21 variations plus a "none" baseline. That's about as controlled a contrast set as you could design by hand, and the "none" condition gives you a neutral anchor: you can measure displacement of each scaffold from baseline, not just Democrat-vs-Republican, which lets you ask whether "I am White" barely moves activations while "I am a Democrat" moves them a lot — directly mirroring their behavioral finding (their Figure 4 shows exactly this pattern in output space for GPT-4o). Supplement with one independent corpus (congressional statements with DW-NOMINATE labels, à la Kim et al.) so your reference ideology direction comes from content, not from scaffolding — otherwise H2's "alignment with the ideology direction" is circular.

Decision 5: The token-identity confound — this one's important. A probe distinguishing "I am a Democrat" prompts from "I am a Republican" prompts might just be reading the literal token "Democrat," which is trivial and uninteresting. Two mitigations: measure at the last token of the question (many tokens after the scaffold prefix, so the signal must have propagated through the model's processing of the whole prompt), and better, test transfer: does the scaffold-trained direction score partisan content (congressional text) correctly, and vice versa? If the scaffolding direction is just a token detector, it won't transfer; if it engages the model's actual ideology representation, it will. This transfer test is arguably your single most important experiment — it's the crux of H2.

Decision 6: Causal evaluation. Steer along the extracted direction during generation on neutral political prompts, sweep the coefficient, judge output slant. Useful details from Kim et al. to borrow: they swept  $\alpha$  in  $\{-30 \dots 30\}$  and found  $|\alpha| > 30$  broke coherence, so expect a similar cliff and report the full dose-response rather than one point; interventions on early-to-middle layers worked while middle-to-late layers did nothing, so intervene where your probe curve peaks, not later; and they validated GPT-4o as a slant judge against politically balanced human annotators with an intraclass correlation of 0.91 — cite that as your license to use an LLM judge in a PoC without running your own annotation study. Controls: random direction of matched norm (should do nothing), and steering on non-political prompts (should leave them untouched — a specificity check).

## FIGURES WORTH INCLUDING

From Cen et al. (arXiv:2509.18446) — your motivation section:

- Figure 4 is your motivating figure. It's a per-model heatmap of pairwise cosine distances between response embeddings under different identity prompt variations — i.e., which user scaffolds produce similar vs. different outputs. It shows GPT-4o reacting most strongly to political-affiliation scaffolds and models implicitly grouping demographics. This is the behavioral phenomenon your entire project proposes a mechanism for, and it sets up a lovely symmetry you should state explicitly in the write-up: they measured scaffold-induced displacement in output-embedding space; you measure scaffold-induced displacement in activation space. Same experimental logic, one level deeper.
- Figure 3 (bar chart of average scaffold sensitivity by model and question category) is the simpler companion if you only want one — it's the "all models are steerable, Gemini most, GPT least" summary.
- Figure 5 or 7 (candidate-adjective associations, and the pre/post-Harris-nomination shift) only if you keep the candidate-framing anecdote in your intro — Figure 5 is the rigorous version of the

"family man vs. patriot" observation.

From Kim et al. (arXiv:2503.02080) — your priors section:

- Figure 2 (their fig:performance): three layer-by-head heatmaps of probe Spearman correlation, visually showing the mid-layer band where ideology is decodable. This is the canonical "linear representation exists and is localized" figure, and it's also the template for the layer-wise plot you'll produce — including it lets a reader compare your residual-stream curve to their head-level map.

- Figure 1: essays with tokens colored blue-to-red by the probe's ideology reading, token by token. The single most intuitive figure in this literature — perfect for explaining "monitoring via probes" to a non-interp audience like a seminar.

- Figure 4 (their fig:intervention, panel a): generated-text slant as a function of steering coefficient alpha across three models — the dose-response template your causal experiment will replicate.

From Arditi et al. (arXiv:2406.11717) — only if your doc spends real space on the refusal analogy: their Figure 2 (ablating the refusal direction bypasses refusal on a harmful prompt) is the iconic one-direction-mediates-behavior illustration. Skippable if refusal stays a one-line aside.

From Kabir et al. (arXiv:2508.21448) — their headline results (7.3x feature count, ablation-increases-refusal) are better summarized in a sentence than reproduced as figures for your purposes; I read their abstract and contributions but didn't pull their figure list, so if you want one, grab whichever plot shows the Gemma-vs-Llama political-feature-count comparison, since it's your justification for choosing Gemma.

On mechanics: for a working doc, screenshot-with-caption-and-citation ("Figure reproduced from Kim et al. 2025") is standard academic practice. Do check licenses if this becomes anything more formal — the arXiv papers carry varying CC licenses (Kim et al. is CC BY-NC-SA; Kabir et al. is CC BY-NC-ND, which is the restrictive one), and when in doubt, linking rather than reproducing is always safe. Also note that Cen et al.'s figure text extractions I have are captions only — pull the actual PDF yourself to grab the images at full resolution.

One last suggestion the Cen paper made me want to add to your design: their embedding-space measurement (MiniLM cosine distances between responses) is cheap enough that you could compute it on your own Gemma generations under the same scaffolds — giving you a behavioral displacement measure and an activation displacement measure on the same model, same prompts. Correlating those two, per scaffold, is about the most direct test of H2 available, and neither paper has done it.

## User

how would i find the meaning behind activated features without lookup

## Claude

There are basically five techniques, in increasing order of effort, and a real interpretation should stack at least two or three of them. This is exactly what Neuronpedia does behind the scenes, so building it yourself also means you understand what the lookup would've told you (and its failure modes).

1. Max-activating examples — the workhorse. Run a large, diverse text corpus through the model + SAE and, for each feature of interest, collect the passages where it fires hardest, keeping the token-level activation values. Then read them: if the top 50 examples are all mentions of tax cuts and deregulation, you have a hypothesis. Practical details that matter: use a general corpus (a

slice of the Pile, FineWeb, or OpenWebText — a few hundred thousand tokens minimum), not just your political prompts, because a feature that looks like "conservatism" on a political-only corpus might actually be "formal legal language" that merely correlates with your data. Also look at examples across the activation distribution — top decile, middle, threshold-level — not just the maximum. Features often look monosemantic at the top and messy in the middle, and the middle is where your actual measurements live. And look at which token the feature fires on within each passage (fires on "abortion" itself vs. fires at the end of stance-taking sentences — very different features).

2. Logit-lens on the decoder direction — free and instant. Take the feature's decoder vector and multiply it by the model's unembedding matrix; the top-promoted and top-suppressed vocabulary tokens tell you what the feature pushes the model to say. A feature whose decoder direction boosts "freedom," "government," "overreach" is telling you something different from one that boosts "she," "her," "women." This takes one matrix multiply, no corpus. Caveat: it works best for later-layer features (whose main effect is on output) and can be misleading for middle-layer features whose effect is mediated by downstream computation — treat it as a hint, not a verdict.

3. Automated interpretation (autointerp) — the scalable version of #1. Feed the max-activating examples (with the firing tokens marked) to an LLM and ask for a short explanation of what the feature detects. This is just the Bills et al. (2023) / Anthropic recipe, and it's ~20 lines of code against the Claude or OpenAI API. The crucial addition that makes it science rather than vibes is scoring: take the generated explanation, show the explainer model new text snippets (a mix of held-out activating and non-activating ones), ask it to predict where the feature fires, and measure agreement with the real activations (detection F1 or correlation). Explanations that don't predict held-out activations get discarded. Without the scoring step, autointerp confabulates plausible-sounding stories at a surprisingly high rate. For a PoC scale — say your top 20-50 candidate features — this costs a few dollars of API calls.

4. Causal steering of the individual feature — the strongest evidence. Clamp the feature to a high value (or ablate it) during generation and read what changes. If clamping feature X makes neutral prompts produce small-government framing, and ablating it removes that framing from Republican-scaffolded generations, you've established what the feature does, which is stronger than what it correlates with. This is more expensive (generation, not just forward passes) so reserve it for the handful of features that survive steps 1-3, but for your project it's non-optional for whichever features you claim carry the scaffolding displacement — it's the same standard Kabir et al. applied with their ablations.

5. Contrast and correlation structure — situates the feature. Compute the feature's activation statistics across your labeled conditions (fires 10x more on Democrat-scaffolded prompts?), its cosine similarity to your probe/difference-in-means directions, and its similarity to other SAE features' decoder vectors (near-duplicate features are common; interpreting one often interprets the cluster). A feature can be perfectly interpretable in isolation ("mentions of religion") and its role in your story only becomes clear from this relational view (high loading on the party displacement direction).

Pitfalls worth one paragraph in your write-up. Max-activating examples suffer from an interpretability illusion — the extremes of a distribution are always more coherent-looking than its bulk, so an explanation fit to the top examples can badly mischaracterize typical firing (this is why scored autointerp on held-out, mid-range examples matters). Interpretations are corpus-relative — a feature is "what it fires on in the data you showed it," so a narrow corpus gives narrow illusions. And some features genuinely are polysemantic or uninterpretable even in good SAEs; the



honest move is to report the fraction of your top displacement-aligned features that passed interpretation scoring, not to force a story onto every one.

Concrete pipeline for your PoC: cache SAE activations over (a) ~500k tokens of general web text and (b) your full scaffolded-prompt set -> for each of the top ~30 features aligned with the displacement direction, extract top-and-mid activating examples -> logit-lens each decoder vector -> autointerp with scoring via the API -> causal clamp/ablate the top ~5 survivors. That's maybe a day of work on top of the infrastructure you'll already have, and it makes your feature-level claims self-contained rather than resting on Neuronpedia's labels — which is also just better methodology to report, since you can state the scoring threshold your interpretations had to clear.

## User

are you sure we shouldn't just use neuronpedias labels

## Claude

No — for a PoC you should use them, and I'd walk back any implication otherwise. My last answer was answering the question you asked ("how would I do it without lookup"), not claiming you should. The right design uses both, with a clear division of labor:

Use Neuronpedia labels for triage. When you project your displacement direction onto 16k+ SAE features and get a ranked list of a few hundred candidates, checking them by hand is exactly what the lookup is for. It's free, instant, and the labels are produced by essentially the same autointerp pipeline I described — someone already paid the API costs. Filtering "which of these 300 features are even plausibly political" via Neuronpedia is a strictly correct use of your time.

Do your own verification only for the features you make claims about. This is the load-bearing distinction. Three reasons the labels alone can't carry your top ~5-10 features:

1. Provenance and granularity. Neuronpedia labels are automated explanations generated from max-activating examples on a general corpus, with variable (sometimes absent) scoring. They're optimized to be roughly right about what a feature does on average web text. Your claims are about fine distinctions on political text — a label like "political terminology" or "US politics" is true and useless for distinguishing whether a feature encodes party identity, policy stance, partisan rhetoric style, or just the word 'Democrat'. Those distinctions are your entire research question, and the generic label genuinely cannot resolve them.
2. Distribution mismatch. The label describes firing behavior on the corpus it was generated from. You care about firing behavior on scaffolded chat prompts, which is a different distribution. The five-minute check — pull your own max-activating examples from your own prompt set for each top feature — frequently reveals that a feature labeled one way behaves more specifically (or differently) in your domain. That check costs almost nothing since you'll have the SAE activations cached anyway.
3. The causal claim needs causal evidence regardless. If your write-up says "scaffolding displaces activations along feature X, which encodes conservative policy framing," no label — Neuronpedia's or your own autointerp's — establishes the second half. Only clamping/ablating X and observing the output change does. This step was in my pipeline not because labels are untrustworthy but because correlational interpretation of any provenance doesn't support causal language.

So the realistic workflow: rank features by displacement alignment -> Neuronpedia lookup to triage hundreds down to a shortlist -> your own activating-examples check on your distribution for the shortlist -> causal clamp for the few you'll actually name in the doc. That's maybe two hours of verification instead of the day-long full pipeline, and in the write-up you cite Neuronpedia labels as the initial identification with a one-line note that you verified top features on your own data. That framing is also what a skeptical reader wants to see — "we trusted the public labels" invites an easy objection; "public labels for search, own verification for claims" closes it.

The full DIY pipeline from my last message becomes relevant only if (a) you use an SAE without Neuronpedia coverage, (b) a feature central to your story has a label that contradicts what you observe, or (c) this graduates from PoC to paper, where reviewers will want scored interpretations. For the weekend version: lookup first, verify narrow, clamp what you claim.

## User

why do i only have to deal with 26 probes instead of 1000

## Claude

It's just counting locations to probe, and it falls out of the architecture.

At each layer, a transformer maintains one running d-dimensional vector per token — the residual stream. Each layer's attention heads and MLP read from it, compute something, and add their outputs back into it. So per token you have two kinds of measurable objects: the one summed stream per layer, or the many individual component outputs per layer.

Residual-stream probing: one vector per layer (at your chosen token position). Gemma-2-2B has 26 layers, so 26 activation sites -> 26 probes. That's where my number came from — it's literally the layer count of the model I recommended.

Attention-head probing (Kim et al.'s approach): each head's output is probed separately, so it's layers x heads. Llama-2-7B has 32 layers x 32 heads = 1,024 probes — that's their number. (If you did head-level probing on Gemma-2-2B, which has 8 attention heads per layer, it'd be  $26 \times 8 = 208$  — smaller, but still ~8x the work, and Gemma Scope's SAEs wouldn't line up with those sites anyway.)

The deeper point isn't the compute — 1,024 ridge regressions is minutes on a laptop, honestly — it's what each choice buys and costs:

- Head-level gives localization: Kim et al. could say "layer 15, head 18 carries ideology," and their steering intervened on the top-K specific heads. The price is a model-selection problem — with 1,024 probes, some will look predictive by chance, so you need cross-validation discipline and multiple-comparison awareness, plus a story for how to ensemble the top heads (their K sweep).
- Stream-level gives sufficiency with simplicity: because every head's output is summed into the stream, any concept linearly decodable from some head is also present in the stream downstream of it (possibly diluted by everything else that got added, which is the real cost — lower resolution, potentially slightly weaker probes). One probe per layer, one obvious peak-layer choice, and direct compatibility with the residual-stream SAEs and the difference-in-means steering recipe.

So "26 vs. 1,024" is shorthand for "one aggregate site per layer in a 26-layer model, versus one site per head in a 32x32 model." For the PoC, the stream is the right resolution; heads are the phase-2 drill-down if you want to say which components write the displacement, not just that it exists.

## User

what do you mean by logistic and ridge probes give me accuracy per layer

## Claude

Concretely, it's this procedure. When you run your prompts through Gemma and cache activations, you get — for each prompt — one vector per layer (the residual stream at your chosen token position). So a dataset of 2,000 prompts gives you 26 separate datasets: {layer-1 vector, label} x 2,000, {layer-2 vector, label} x 2,000, and so on, where the label is the thing you care about (Democrat vs. Republican scaffold, or a DW-NOMINATE score).

For each layer independently, you train a small supervised model — a probe — to predict the label from that layer's vectors, and evaluate it on held-out prompts. That yields one number per layer: how well can the label be linearly read out of the representation at that depth. Plot those 26 numbers against layer index and you have the "accuracy per layer" curve.

The two probe types just match the two label types:

- Logistic regression for categorical labels (Democrat/Republican scaffold -> binary classification). The per-layer number is held-out classification accuracy or AUC. This is what you'd use on the Cen-derived matched pairs.
- Ridge regression for continuous labels (DW-NOMINATE ideology scores, which run -1 to +1). The per-layer number is held-out Spearman correlation or R-squared. This is what Kim et al. used (ridge with  $\lambda=1$ , 2-fold CV, Spearman as the metric) — the same recipe, just at head-level sites instead of layer-level.

Both are linear models on purpose: the question isn't "can anything decode party from activations" (a big MLP probe can decode almost anything, which proves little) but "is party represented as a linear direction," which is the representational claim from the literature you're testing. Kim et al.'s robustness check — nonlinear probes didn't beat linear ones — is what licenses stopping at linear.

What the curve buys you, in order of importance:

1. Existence check. If no layer beats chance, H1 is dead before you touch SAEs or steering, and you found out in an afternoon.
2. Layer selection. The curve typically rises from near-chance in early layers, peaks somewhere in the middle, and often flattens or dips late. The peak tells you where to point everything downstream: which layer's Gemma Scope SAE to load, where to compute difference-in-means, where to intervene. (This expectation isn't folklore — Kim et al. found the signal at layers ~10-20, and their steering worked at early-to-middle layers and did nothing at late ones.)
3. A shape that's itself evidence. Signal appearing only in middle layers is the signature of a computed, abstract representation; a curve that's high from layer 1 would suggest you're reading surface token identity (the word "Democrat" sitting in the embedding) rather than anything the model inferred about the user — which loops back to the token-identity confound. The rise of the curve across depth is part of your argument that scaffolding engages real processing.

Mechanically it's trivial once activations are cached: a for-loop over 26 layers, LogisticRegression or Ridge from sklearn inside, cross-validated score out, one matplotlib line plot. Seconds of compute. The care goes into the split, not the fit — hold out at the level of base questions, not prompt rows, so the same question never appears in both train and test under different scaffolds; otherwise the probe can partly memorize question content and your accuracy is inflated.

And to tie it to the earlier thread: this per-layer curve is your Figure 1, the direct analog of Kim et al.'s layer-by-head heatmap collapsed to one line — theirs had two axes because they probed every head separately; yours has one because you probe the summed stream.

## User

how does ridge regression workj

## Claude

Start with plain linear regression and add one twist.

Ordinary least squares. You have inputs  $x$  (for you: a 2,304-dimensional activation vector from one layer of Gemma-2-2B) and a target  $y$  (a DW-NOMINATE score). You assume the relationship  $\hat{y} = \theta^T x$  and pick the weight vector  $\theta$  minimizing squared error over your  $N$  training examples:

$$L(\theta) = \sum_i (y_i - \theta^T x_i)^2$$

The problem in your regime. OLS breaks down exactly in the situation probing puts you in: dimensionality comparable to or exceeding the sample count (Kim et al. had 552 lawmakers; you might have a few thousand prompts against 2,304 dimensions), and heavily correlated input coordinates (activation dimensions are nothing like independent). Two consequences. With  $d \geq N$ , infinitely many  $\theta$  fit the training data perfectly — the system is underdetermined — and OLS has no way to prefer one, so it happily memorizes noise. And with correlated coordinates (multicollinearity), even when a unique solution exists it's wildly unstable: if two activation dimensions carry nearly the same signal, OLS can assign +1000 to one and -999 to the other, and tiny changes in training data swing the weights enormously. Held-out performance craters either way.

The ridge fix: penalize weight size.

$$L_{\lambda}(\theta) = \sum_i (y_i - \theta^T x_i)^2 + \lambda \|\theta\|^2$$

That second term charges the model for the squared magnitude of its weights. Now the optimizer trades off fit against magnitude: among all the solutions that explain the data about equally well, it picks the smallest one.  $\lambda$  sets the exchange rate —  $\lambda = 0$  recovers OLS;  $\lambda \rightarrow \infty$  shrinks all weights toward zero (predicting the mean); in between you get solutions that capture real signal but can't afford the enormous cancelling weights that noise-memorization and multicollinearity produce. On the correlated-pair example: instead of +1000/-999, ridge spreads modest weight across both, which is both stabler and — for your purposes — a more honest "direction."

Two useful facts. It has a closed-form solution:  $\theta = (X^T X + \lambda I)^{-1} X^T y$ . The  $\lambda I$  added to  $X^T X$  is literally "the ridge" — it makes the matrix invertible even when  $X^T X$  is singular (which it always is when  $d > N$ ), so the underdetermination problem disappears by construction. That's why sklearn's Ridge fits in milliseconds: one matrix solve, no iteration. And it's the regression counterpart of the L2 penalty in the logistic regression you'd use for the binary labels — sklearn's LogisticRegression applies the same shrinkage by default (its  $C$  is  $1/\lambda$ ) — so both of your probe types are "linear model + L2," differing only in loss function.

Why it's the standard probing choice specifically. Beyond making the fit well-posed, the probing literature cares that the learned  $\theta$  is the deliverable: it's your candidate ideology direction,

reused as a steering vector (Kim et al. add  $\alpha * \sigma\text{-hat} * \theta\text{-hat}$  to activations) and compared by cosine to difference-in-means and SAE decoder vectors. An unstable, noise-shaped  $\theta$  would poison everything downstream; the shrunken, stabilized ridge  $\theta$  is a usable geometric object. Lasso (L1 penalty) is the alternative you might wonder about — it zeroes out coordinates entirely, giving sparsity — but that's the wrong prior here, since there's no reason a concept direction should align with a few individual neuron axes; distributed small weights are exactly what the linear-representation picture expects. (If you want sparsity in an interpretable basis, that's what the SAE step is for.)

Choosing  $\lambda$ . Cross-validation: fit at several values, keep the one with the best held-out score. Kim et al. swept  $\{0, 0.001, 0.01, 0.1, 1, 100, 1000\}$  with 2-fold CV and landed on  $\lambda = 1$ ; the standard result is a U-shaped validation curve — underfit at huge  $\lambda$ , overfit at tiny  $\lambda$  — that's typically flat near its bottom, so precision doesn't matter much. RidgeCV does the sweep for you in one call. One practical footnote: standardize your activation features first (ridge penalizes all coordinates equally, so it's scale-sensitive), and remember your held-out split should be by base question, per the earlier point —  $\lambda$  tuning inherits the same leakage rules as everything else.

## User

what is output slant

## Claude

"Output slant" is just the political lean of the text the model generates — measured after the fact, from the words alone, the way you'd rate an op-ed as liberal-leaning or conservative-leaning without knowing anything about the author. I've been using it to name the behavioral, output-side variable, in deliberate contrast to everything else in your pipeline, which is internal: probe directions, displacements, and feature activations live in activation space; slant lives in the generated text.

It matters because it's the dependent variable of your causal experiment. The steering claim has the form "intervene on activations  $\rightarrow$  generated text becomes measurably more liberal/conservative," and the second half needs a number. So the dose-response plot you're aiming for has steering coefficient  $\alpha$  on the x-axis and measured output slant on the y-axis; "slant shifts monotonically with  $\alpha$ " is the result that establishes your direction as causal rather than merely correlational.

Operationally, you get the number by scoring each generated passage on a scale. The options, roughly in order of convenience:

1. LLM-as-judge — show each generation to a strong model with a rubric like "rate this text's political perspective from 1 (strongly conservative) to 7 (strongly liberal)." This is what Kim et al. did: GPT-4o rated their 1,134 steered essays on that 7-point scale, and they validated it against ten politically balanced human annotators (3 Democrats, 4 independents, 3 Republicans), getting an intraclass correlation of 0.91 — which is your citable license to use a judge without running your own annotation study.
2. Zero-shot stance classifier — an off-the-shelf NLI/stance model from HuggingFace scoring agreement with liberal vs. conservative position statements. Cheaper and fully reproducible, but coarser.
3. Your own probe, applied to the generation — run the generated text back through Gemma and read the ideology probe's output on it, like Kim et al.'s token-coloring figure. Elegant, and fine as a

supplementary measure, but it can't be your primary one for the steering experiment: you'd be validating the intervention with the same internal direction you intervened on, which is circular. The judge (or classifier) is external to the machinery being tested, which is what makes it evidence.

A useful precision for the write-up: slant is a property of a passage, not of the model — the same model produces different slants under different scaffolds or steering, which is the entire phenomenon. And it's distinct from two neighbors it's easy to blur with: model bias (the average slant at  $\alpha = 0$ , no intervention — Kim et al. found all three of their models sat slightly liberal-of-center unsteered, which you'll likely replicate as a baseline observation) and refusal/hedging (a generation can dodge taking any stance; worth tracking as a separate outcome rather than forcing the judge to place evasive text on the liberal-conservative axis — give the judge an explicit "no discernible stance" option so evasion doesn't masquerade as centrism).

## User

what do you mean by sweep the coefficient

## Claude

When you steer, you add your extracted direction  $v$  into the residual stream during generation:

```
r <- r + alpha * v
```

Alpha (the coefficient) is a scalar knob controlling how hard you push, and its sign controls which way — negative alpha steers liberal, positive steers conservative (or vice versa, depending on how you oriented  $v$ ).

"Sweeping" just means: don't run the experiment at one value of alpha — run the same generation experiment at a whole grid of values, e.g. alpha in  $\{-30, -20, -10, 0, +10, +20, +30\}$  (Kim et al.'s exact grid), holding everything else fixed — same prompts, same layer, same seed regime. Each alpha produces a batch of generations; you score each batch's output slant; then you plot slant ( $y$ ) against alpha ( $x$ ). That plot is the dose-response curve, on direct analogy with pharmacology: dose of the intervention on one axis, size of the response on the other.

Why this is worth  $\sim 7\times$  the compute of a single run:

1. One point is uninterpretable; a curve is evidence. Suppose you only ran  $\alpha = +20$  and slant came out more conservative. Fluke? Prompt sensitivity? Generic text degradation the judge misreads as slant? You can't tell. But if slant increases monotonically as alpha goes  $-30 \rightarrow +30$ , passing through your unsteered baseline at  $\alpha = 0$ , that graded, ordered relationship is very hard to produce by accident — it's the signature of a genuine causal knob. Monotonicity is the result; a single point can't show it.
2. You don't know the right magnitude in advance. Activation norms vary by model and layer, so a useful alpha for Gemma-2-2B at layer 13 is not something you can predict from Kim et al.'s Llama numbers. Too small and nothing happens (false null); too large and you hit the coherence cliff — Kim et al. found  $|\alpha| > 30$  produced incoherent text. The sweep locates the usable window empirically instead of by guessing.
3. It exposes the fluency trade-off honestly. Steering degrades text quality as magnitude grows. Reporting the whole curve, ideally with a fluency measure (perplexity or a judge's coherence rating) alongside slant, shows how much behavioral shift you get before the model breaks — versus a single cherry-picked alpha, which invites the objection that you found the one setting

that works.

4. Your control condition inherits the sweep. The random-direction control (same-norm random vector) should be run over the same alpha grid. The contrast you want is "the ideology direction produces a sloped dose-response; a random direction produces a flat line until both hit the cliff" — flat-vs-sloped over the same grid is a much cleaner comparison than two single points.

Mechanically it's a nested loop — for alpha in grid: for prompt in prompts: generate with hook adding  $\alpha \cdot v$ ; judge outputs — and it's your longest-running job because it's generation rather than single forward passes (7 alpha values x ~50 prompts x ~150 tokens each, per direction tested). This is the plot I referred to earlier as your steering deliverable: Kim et al.'s Figure 4a is precisely slant-vs-alpha for three models, and your version of it — plus the flat control line — is the causal centerpiece of the PoC.

One design detail: apply the same  $\alpha \cdot v$  at every generated token position (standard practice, what Kim et al. did) rather than only at the prompt's last token, so the push is sustained through the whole generation; and normalize  $v$  to unit norm before sweeping so alpha has a consistent meaning across the layers and directions you compare.

## User

*[The user attached the full working document, v3 — "User-Identity Induced Political Viewpoints" draft with The Idea, Priors, hypotheses, and Methodology sections (Model, Probe Target, Direction extraction, Data, Causal Eval).]*

hows this and where should i add the cosine similarity thing and what exactly should i add:

## Claude

This is in good shape — the gap statement landed, the confound section is the strongest part of the doc, and the reasoning-per-decision format reads well. Fixes first, then the cosine similarity placement with paste-ready text.

Errors to fix:

1. "No need for cross-validation" (Probe Target, reason 3) is wrong and will get flagged by any reader who knows the area. You absolutely still cross-validate: held-out evaluation is what makes the per-layer accuracy numbers meaningful, and lambda for ridge is chosen by CV. What the residual stream spares you is the head-selection machinery — 208 probes creating a multiple-comparisons problem and requiring top-K ensembling logic like Kim et al.'s K sweep. Rewrite as: "No head-selection machinery: probing 208 sites creates a multiple-comparisons problem and requires ensembling logic (cf. Kim et al.'s top-K head selection); one probe per layer avoids this. Standard cross-validation still applies for probe evaluation and regularization tuning."
2. Gemma reason 2 has the logic backwards. You wrote a negative result is "less likely to be attributable to weak political representations than to a failed experiment" — which says a null means you messed up, undermining your own design. The point is the opposite: because Gemma's political features are known to be rich, a null result is informative — it means scaffolding doesn't engage those features, rather than the features not existing. Rewrite: "...a null result on Gemma is therefore interpretable: since its political features are known to exist and be manipulable, a failure to find scaffold-induced displacement would indicate that scaffolding does not engage those representations, rather than that the representations are absent."

3. H2 is still the untestable version. "Predicts the behavioral differences found by Cen et al." — their behavioral data comes from closed models you can't get activations from. Split it: "H2a: the scaffold-induced displacement direction aligns (high cosine similarity) with an independently extracted, content-derived ideology direction. H2b: steering along the displacement direction shifts output slant in the direction predicted by the scaffold, reproducing in an open model the kind of framing sensitivity Cen et al. observed behaviorally."

4. Data section is missing the second corpus. Your confound-mitigation bullet says you'll "score partisan text (like the data used by existing literature)" but the Data section only lists the Cen dataset. Add explicitly: a content-labeled corpus (e.g., congressional statements with DW-NOMINATE scores, following Kim et al.) serving two roles — the source of the independent reference ideology direction for H2a, and the transfer test set for the token-detector confound. Without naming it, H2a has no reference direction to compare against.

5. Causal Eval is missing its controls and the sweep structure. "At different coefficient scales" should become the explicit protocol: normalize the direction to unit norm; sweep alpha over a symmetric grid (Kim et al. used  $\{-30 \dots +30\}$  and found incoherence beyond  $|\alpha| > 30$ ); report the full slant-vs-alpha dose-response curve, where monotonicity through the  $\alpha=0$  baseline is the causal evidence; run a matched-norm random direction over the same grid as a control (expected: flat); and generate on non-political prompts as a specificity check (expected: unaffected). Also give the judge a "no discernible stance" option so refusals/hedging don't score as centrism.

6. Small: "Cen et al.." (double period, twice), "a la" -> "à la", and Gurgurov et al. 2025 — I haven't verified that's the right author name for the multilingual steering paper (arXiv:2507.22623); double-check it matches the actual author list before it fossilizes in the doc.

Where the cosine similarity goes — three places, two of which you have or half-have:

Place 1 (you already have it): Direction extraction -> Linear probes bullet — probe-normal vs. difference-in-means cosine as internal corroboration. Keep as is. One wording fix: for a probe, the weight vector IS the normal to the decision hyperplane, so "find the vector orthogonal to the hyperplane" can just be "use the probe's weight vector as a direction estimate."

Place 2 (the main one — this is "the cosine similarity thing"): You need an Analysis (or "Hypothesis tests") subsection between Causal Eval and Results, because right now the doc describes how you extract directions but never states the measurement that decides H1/H2a. Paste something like:

"Analysis. For H1: compute the displacement vector for each scaffold condition (mean activation under scaffold minus mean under the 'none' baseline, per layer, at the probe-selected layer) and test whether its magnitude exceeds a null distribution built from random splits of baseline prompts. For H2a: compute the cosine similarity between the Democrat-Republican scaffold displacement direction and the reference ideology direction extracted from the content-labeled corpus. Because random vectors in  $\sim 2,300$  dimensions have cosine similarity concentrated near zero, I'll report the observed cosine against an empirical null (cosines of matched random directions, and the cosines between the ideology direction and non-political scaffold displacements, e.g. 'I am White' minus baseline) rather than against an arbitrary threshold. A high political-scaffold cosine alongside near-zero non-political-scaffold cosines supports H2a; systematic displacement (H1 holding) with near-zero cosine to the ideology direction would be the informative null — scaffolding moves activations along persona directions orthogonal to ideology."

That last sentence matters: it pre-registers what a negative result means, which makes the write-up credible either way.



Place 3 (optional but the most novel measurement available to you): at the end of Analysis or in Next Steps — correlate internal and behavioral displacement on the same model: for each scaffold, compute both the activation-space displacement (projection onto the ideology direction) and the output-embedding displacement of Gemma's own generations under that scaffold (Cen et al.'s MiniLM cosine-distance measure, applied to your generations). A per-scaffold correlation between the two is the most direct internal-explains-behavioral evidence you can produce, and neither paper has done it. If it's too much for the PoC, park it in Next Steps explicitly — it's a strong "phase 2" flag for a reader.

One structural note to close: your intro's third sentence still frames the question as "along identifiable directions... or if different scaffolds map to different directions" — that's an H1-flavored phrasing, and now that the doc ends with formal hypotheses, consider making the intro sentence mirror H1/H2 exactly so a reader hits the same claim twice in the same words. Consistency between the prose question and the numbered hypotheses is cheap polish that makes the whole doc feel pre-planned rather than assembled.

## User

i feel like i am white could also be politically load-bearing though so what would an actual neutrality look like

## Claude

You're right, and that's a flaw in my suggested control, not a nitpick. In US politics essentially every demographic category carries partisan information — white voters lean Republican in aggregate, Black voters lean heavily Democratic, gender has a persistent gap, age, religion, education, even urban/rural residence all load onto the partisan axis. A model that's absorbed US political discourse SHOULD treat "I am White" as weakly politically informative, and Cen et al.'s own Figure 4 finding — that models implicitly group demographics and treat some as more "representative of the general population" — is behavioral evidence that they do. So "political scaffold projects onto ideology, demographic scaffold doesn't" was always going to be a strawman dichotomy: the honest expectation is a spectrum of loadings, not a clean orthogonal class.

That gives you two moves, and the second is better than what it replaces.

Move 1: build genuinely inert controls. If you want scaffolds that are as close to politically null as anything gets, they have to be non-demographic and non-lifestyle: "I am left-handed," "I was born in June," "I am an only child," "My favorite color is blue." (Be careful with seemingly innocuous lifestyle traits — vegetarianism, gun ownership, pickup trucks, even dog-vs-cat preference carry measurable partisan correlation in survey data; birth month and handedness don't.) These aren't in the Cen prompt suite, so you'd generate a small supplemental set yourself — same prefix format, same base questions, maybe 4-5 inert identities — which costs an hour. Alongside these, add syntactic controls that strip semantics entirely while preserving form: "I am a person," "My name is Alex" — these control for the possibility that ANY first-person prefix displaces activations somewhat, separating "prefix effect" from "identity effect." And add a paraphrase consistency check: "I am a Democrat" and "I'm a member of the Democratic party" should displace in nearly the same direction; if they don't, your directions are surface-form artifacts.

Move 2 — the better one: stop treating neutrality as an assumption and make the loading spectrum the measurement. Instead of a binary political-vs-neutral contrast, compute the projection of EVERY scaffold's displacement onto the ideology direction and rank them. The prediction is now graded: explicit partisan scaffolds ("I am a Democrat/Republican") at the

extremes, strongly partisan-correlated demographics ("I am Black") next, weakly correlated ones ("I am a woman," "I am White") closer to zero, inert scaffolds ("I am left-handed") and syntactic controls at zero. That ordering is a much richer and more falsifiable prediction than orthogonality — and here's the payoff: you can quantify the expected ordering externally. Pew's validated-voter data gives you each demographic's actual partisan lean (Black voters ~85-90% Democratic, white evangelicals ~75-80% Republican, women D+10-ish, etc.). Correlate each scaffold's projection magnitude against its real-world partisan lean. A significant correlation says the model's internal political loading of user identities tracks empirical American voting patterns — which is simultaneously your H2a evidence, a mechanistic account of Cen et al.'s demographic-grouping result, and frankly the most publishable-flavored single plot the PoC could produce. The stereotype observation from the talk ("family man" for women users) becomes a quantitative claim: the model maps demographic identity onto the political axis in proportion to real-world demographic partisanship.

Your nulls then come from constructions that need no semantic assumptions at all: a permutation null (shuffle scaffold labels across matched pairs, recompute displacements, repeat to get the distribution of projections under "no real identity effect") and the matched-norm random-direction null you already have. Those replace "I am White should project to zero" as the statistical baseline, so nothing in your inference rests on declaring any human category apolitical.

Concretely, edit the Analysis passage I gave you: delete the clause using non-political demographic scaffolds as the near-zero comparison, and replace it with something like — "Rather than assuming any scaffold is politically neutral, I compute each scaffold's displacement projection onto the ideology direction and evaluate (a) the full ranking against inert and syntactic controls, with statistical baselines from permutation and random-direction nulls, and (b) the correlation between demographic scaffolds' projection magnitudes and those demographics' real-world partisan lean (per validated voter surveys). H2a is supported if partisan scaffolds project strongly, controls project at null levels, and demographic projections track empirical partisanship; H1-without-H2a (systematic displacement orthogonal to ideology) remains the informative null."

One honest caveat to carry into the doc: the Pew-correlation analysis has few data points (however many demographic scaffolds the Cen suite contains, likely under ten), so report it as a rank correlation with appropriate humility rather than a headline statistic — it's a compelling consistency check at PoC scale, and a properly powered version (many more identity scaffolds) is a natural Next Steps item.

## User

*[The user attached the working document, v4 — now including labeled H2a/H2b analysis with the spectrum framing, permutation/random-direction nulls, Pew correlation, and an expanded Causal Eval.]*

hows this:

## Claude

This is close to done — the H2a analysis section now contains the intellectual core of the project, and the hypothesis structure with sub-parts works. Remaining items, in priority order:

1. The H2a paragraph buries its insight in narrative. "Initially, I was going to... However, it was hard for me... Then I realized..." — the honesty is good, but the key sentence is tangled: "it was hard to come up with scaffolding that 100% aligned me to a policy, unless the identity was that political belief." What you mean is cleaner than what's written. Suggested rewrite that keeps the reasoning but sharpens it:

"An initial design compared the scaffold direction against 'politically neutral' demographic controls, but on inspection no demographic identity is politically inert — in US politics, essentially every demographic category (race, gender, religion, age) carries partisan information, and only explicit political affiliation aligns directly with the ideology axis. Rather than assume neutrality anywhere, I treat political loading as a spectrum to be measured: I compute each identity scaffold's displacement projection onto the ideology direction and evaluate..."

Then your two bullets follow. This turns a "I struggled with controls" story into a design contribution, which is what it actually is.

2. Data section doesn't mention creating the control prompts. Your Analysis references inert/syntactic controls ("I was born in June," "I am a person"), but those aren't in the Cen suite — you have to generate them. Add one line to Data: "I'll supplement the Cen-derived pairs with a small self-generated set of inert scaffolds (birth month, handedness, 'I am a person') in the same prefix format over the same base questions, since the released suite contains no politically inert conditions." Also worth adding the paraphrase check here ("I am a Democrat" vs. "I'm a member of the Democratic party" should displace near-identically) — it's a cheap surface-form robustness test that fits naturally next to the token-confound bullets.

3. Causal Eval is missing its controls — this is the biggest substantive gap left. Right now it reads as "sweep alpha, judge slant," but the sweep alone doesn't isolate causation. Add: (a) normalize the direction to unit norm so alpha is comparable across directions/layers; (b) a matched-norm random direction run over the same alpha grid — expected flat, and flat-vs-sloped over the same grid is the causal contrast; (c) a specificity check: steer during generation on non-political prompts, expected unaffected; (d) track fluency/coherence alongside slant so the trade-off is visible. Also note that Kim et al.'s  $\{-30...+30\}$  window is THEIR calibration for Llama-scale activations — your usable range on Gemma-2-2B will differ, so treat their grid as a starting point and locate your own coherence cliff empirically. Finally, state H2b's actual prediction explicitly: slant shifts monotonically with alpha, in the sign predicted by the scaffold that generated the direction — the sign prediction is what ties the causal result back to scaffolding rather than just "some direction steers politics."

4. H2a bullet 2 needs its data source and caveat. Name where real-world partisan lean comes from (e.g., Pew validated-voter data) and flag the small-n honestly — you'll have fewer than ten demographic scaffolds, so this is a rank correlation reported as a consistency check, with a properly powered version (many more identity scaffolds) as a Next Steps item.

5. Pre-fill the empty skeletons. Even for a working doc, listing the planned figures under Results reads as pre-registration and keeps you honest: (1) probe accuracy vs. layer; (2) per-scaffold projection ranking against controls and nulls; (3) projection-vs-real-world-partisanship scatter; (4) SAE feature loading of the displacement; (5) slant-vs-alpha dose-response with random-direction control. Under Next Steps, park the phase-2 items already discussed: head-level attribution of which components write the displacement; confirmation on Gemma-2-9B; and the internal-vs-behavioral correlation on the same model (per-scaffold activation displacement vs. output-embedding displacement of Gemma's own generations, Cen-style) — that last one is the most novel measurement available and deserves an explicit flag even if it's out of PoC scope.

6. Small fixes. "Cen et al.." double period (twice); "coefficient scales ." stray space; "Gemma 2-2B" vs "Gemma-2-2B" inconsistency; and the Kim figure lead-in — the heatmap shows probe performance across ALL heads (that's the point: the mid-layer band), so "showcasing the performance of a predictive attention-head" -> "showcasing probe performance across all attention heads and layers, with predictive heads concentrated in middle layers." Also H2's parent

line still says "predicts the behavioral differences found by Cen et al." — with H2a/H2b underneath it now functions as an umbrella, but soften "predicts" to "is consistent with / offers a mechanism for," since prediction of their closed-model measurements isn't literally testable.

One genuine compliment to end on: the Neuronpedia bullet's parenthetical "(which would be necessary eventually for the project)" and the H2a redesign both show the doc doing the thing good methods sections do — recording not just what you'll do but what you consciously deferred and why. Keep that register through Results when the numbers come in.

## User

what do you mean by specificity check - is there such a thing as a non-political prompt or a prompt where politics cannot be injected

## Claude

You're applying the same (correct) skepticism from the scaffold-neutrality discussion, and the answer has the same shape: no, there's no prompt that politics CANNOT be injected into — push alpha hard enough and a banana bread recipe will start editorializing about seed oils and gas stoves, both of which are, in fact, actual culture-war topics, which rather proves your point. Any topic can be politicized because the model has learned a world where most topics have been. If the specificity check required politically sterile prompts, it would be unrunnable.

But that's not what it requires, because the check is a relative measurement, not an absolute one. The claim being tested isn't "this direction cannot affect non-political text" — it's "this direction's effect is concentrated on political content rather than being a generic text-mangler."

Operationally: at the alpha values where political prompts show strong slant shifts, run the same steering on maximally distant prompts (math word problems, cooking instructions, code documentation, a description of photosynthesis) and measure two things — does injected political content appear (topic-intrusion rate, which your judge can score: "does this response contain political framing unrelated to the request?"), and does text quality degrade. The informative result is the asymmetry: large on-target effect, small off-target effect at matched dose. Some off-target leakage at high alpha is expected and honest to report — the refusal-direction literature found exactly this kind of imperfect selectivity, and Arditi et al.'s whole point about "minimal effect on other capabilities" was a measured claim, not an assumed one.

Why the check matters at all: without it, a skeptic can say your "ideology direction" is really a "produce opinionated/emotive/verbose text" direction that merely manifests as slant when the topic happens to be political — the judge would score that as dose-responsive slant on political prompts, and you'd never know the difference. The off-target condition discriminates between these: a genuine ideology direction leaves the math problem alone (or degrades it generically at high alpha, same as the random control does); a "be opinionated" direction makes the math answer preachy. So the thing being tested for specificity is the DIRECTION, not the prompts — the prompts just need to be far enough from politics that intrusion is conspicuous when it happens.

And you can import your spectrum move wholesale rather than pretending a binary: pick off-target prompts at graded political distance — arithmetic (maximally distant), photosynthesis (distant), a recipe (mostly distant, with landmines like the gas-stove example), a question about cities (closer — urbanism is coded), a question about universities (closer still). If intrusion rate rises with the prompt's political adjacency, that's not a failed control; it's an extra, coherent finding — the direction activates in proportion to how politically chargeable the context is, which mirrors exactly what you did with scaffold loadings. Then your write-up can say the same

sentence in both places: nothing is assumed neutral; neutrality is measured as position on a spectrum, and the prediction is about the gradient, not about zeros.

One practical note for the doc: this means the Causal Eval needs a second judged quantity beyond slant — an intrusion/coherence score for off-target generations — and the deliverable becomes a small 2x2-ish comparison: {ideology direction, random direction} x {political prompts, distant prompts} across the alpha grid. Ideology-on-political is the only cell that should show a steep dose-response; that pattern, rather than any single curve, is the specificity result.

## User

*[The user attached a screenshot of Kim et al.'s Figure 1 — the token-colored immigration and abortion essays.]*

also for the kim one im using this

## Claude

Good choice — that's the figure I'd have picked too if you're only including one from Kim et al. It's the most immediately legible artifact in the paper for a non-interp reader: blue/red tokens make "the model's internals carry a readable ideology signal during generation" obvious without any methodology exposition. Three adjustments to how it sits in your doc:

1. Your lead-in sentence needs updating to match this figure. The draft says "Kim et al.'s figure showcasing the performance of a predictive attention-head" — with this figure that's almost right (it genuinely is one head, Layer 16, Head 1), but "performance" suggests an accuracy plot, and what the figure actually shows is monitoring: a probe trained only on lawmaker names reading interpretable token-by-token slant off open-ended essays. Something like: "Kim et al. demonstrate this direction is readable during generation — a probe trained to predict lawmakers' DW-NOMINATE scores annotates open-ended essay tokens with interpretable liberal/conservative slant (their Figure 1, reproduced below):"

2. Be aware of what this figure does and doesn't evidence in your priors paragraph. Your surrounding text makes two claims: (a) linear representation localized to middle layers, and (b) similar viewpoints cluster. Figure 1 directly supports neither — it supports a third claim you should get credit for: the probe generalizes beyond its training distribution (names -> generated policy text), which is the qualitative twin of your own planned transfer test for the token-identity confound. Worth one sentence making that connection explicit, since it pre-legitimizes your Data section's second mitigation: "This generalization from lawmaker-name prompts to open-ended text is the same transfer logic I apply to distinguish an ideology direction from a token detector." The mid-layer localization claim (a) is then carried by citation alone — which is fine for a working doc; if you later want it carried visually, that's the layer-by-head heatmap (their Figure 2), where the dark band across layers ~10-20 IS the localization result. (Note the pleasant consistency: Layer 16 of 32 in this figure sits exactly in that band — you can point that out in one clause.)

3. Caption it as a reproduction. Under the image: "Figure reproduced from Kim, Evans & Schein (2025), Figure 1. Tokens colored by the slant predicted from a single attention head (Layer 16, Head 1) of Mistral-7b-instruct; blue = liberal, red = conservative." Their arXiv version is CC BY-NC-SA, so reproduction with attribution in a non-commercial research doc is squarely fine — the share-alike clause only bites if you're licensing your own doc restrictively, which for a working PoC document isn't a live concern.

One small bonus this figure buys you: it's the precedent for option 3 in your slant-measurement toolkit (running generations back through your own probe for token-level visualization). If your PoC works, producing THIS EXACT FIGURE on Gemma with your scaffold-derived direction — same essay prompts, your direction doing the coloring — would be the single most communicative results figure you could make, and a direct visual answer to "does the scaffold direction read the same signal their content direction reads." Might be worth a line in your planned-figures list.

## User

should we also do token by token

## Claude

Yes — and for your project it's actually more than a visualization nicety, because token-by-token analysis comes in three flavors with very different value-to-cost for you:

1. Positional displacement profile on the PROMPT — do this, it's core. This is the version neither Kim et al. nor anyone else in your citation list has done, and it directly upgrades your token-identity-confound section. Since a single forward pass gives you activations at EVERY position anyway, compute the projection onto your ideology direction at each token position of the scaffolded prompt: across the scaffold prefix ("I am a Democrat"), through the question tokens, to the final token. The two competing hypotheses make different positional predictions. If the direction is a token detector, projection spikes at "Democrat" and decays as the question proceeds — the signal is ABOUT THE WORD. If the model genuinely conditions on user identity, the displacement persists (or re-emerges at semantically relevant question tokens) all the way to the last position — the signal is CARRIED STATE about the user. Right now your mitigation is "measure at the last token and hope the signal propagated"; this analysis replaces the hope with a measurement, and the persist-vs-decay profile is arguably a headline figure. Cost: nearly free in compute, but it changes your caching plan — you'd been implicitly caching one vector per prompt per layer; full-position caching for 2k prompts x ~50 tokens x 26 layers x 2,304 dims lands around 10-15 GB on disk. Fine, but decide up front; or cache all positions only at your 3-4 peak layers.

2. Token-colored generations (the Kim Figure 1 move) — do it if steering works. Run your steered/scaffolded generations back through the model, project each generated token onto your direction, color blue-to-red. As I said last message, reproducing their figure with your scaffold-derived direction is the most communicative results artifact available — it shows a seminar audience in one glance that the scaffold direction reads the same liberal/conservative signal their content-derived direction reads. But it's qualitative and comes after the quantitative results exist; slot it as a results-presentation item, not a methodology dependency. (Kim et al. themselves flag their version as qualitative and leave systematic evaluation to future work — echo that framing.)

3. Token-level SAE feature analysis — explicitly defer. Asking "which SAE features fire at which tokens of the scaffold and question" is a real phase-2 question (it's where head-level attribution and feature-level dynamics live), but it multiplies your interpretation workload by sequence length and adds nothing to H1/H2 as stated. One line in Next Steps.

Where it goes in the doc: add the positional profile as a third bullet in your token-confound section — "(3) I compute the displacement projection at every prompt position, distinguishing a decaying token-locked signal (token detector) from a persistent one (carried user representation)" — and add "positional displacement profile" to the planned-figures list. Then update the caching sentence in Data to say all-position caching at selected layers. The generation-coloring figure goes

in planned figures with a conditional flag ("if H2b holds"), and the SAE-token analysis goes to Next Steps.

One methodological footnote for flavor: if you see the projection GROW through the question tokens — the model becoming progressively "more sure" about the user's politics as it processes a political question in light of the identity prefix — that's an in-context inference dynamic worth calling out explicitly, and it's the kind of observation that turns a PoC into a talk slide.

## User

lets sort this out are we sure we need the positional profile here and if so can you explain it clearly. are you sure we shouldn't just do the token colored generations as a visual thing

## Claude

Fair push — let me be honest about where I overstated. You don't NEED the positional profile. Let me lay out what it is, what it would actually buy, and then the verdict.

What the positional profile is, plainly. A prompt like "I am a Democrat. What is your view on healthcare policy?" gets processed token by token, and a forward pass gives you a residual-stream vector at every position, not just the last one. The profile is: project each position's vector onto your ideology direction and plot projection vs. position. You'd expect a spike at "Democrat" (the word itself is obviously partisan-loaded). The question is what happens after: does the projection decay back toward baseline as the question tokens are processed (signal was just the word sitting in context), or does it persist through to the end (the model is carrying "this user is a Democrat" as state)? Persist-vs-decay is the diagnostic.

Why I called it core, and why that was too strong. I framed it as upgrading your token-confound defense. But walk the logic: your TRANSFER TEST already settles that question decisively. If the scaffold-trained direction correctly scores congressional statements — text containing no "I am a..." prefix at all — then the direction cannot be a mere prefix-token detector; there's no prefix there to detect. Transfer is the inferential kill-shot. The positional profile, by contrast, is descriptive: it tells you the within-prompt dynamics of the signal, which is interesting, but it doesn't rule anything in or out that transfer doesn't already handle. So it's a diagnostic garnish on an argument that's already closed.

And it's less free than I implied. Two practical warts I glossed over: raw projections at different positions aren't directly comparable (activation norms and statistics drift systematically across sequence position — early-token and attention-sink weirdness is well documented), so you'd need to compare scaffolded vs. baseline prompts at matched positions — except the baseline lacks the prefix, so positions are offset by the prefix length and you're aligning on question tokens with an index shift. All solvable, but it's a half-day of careful bookkeeping plus ~10x the activation cache, for a figure that decorates rather than decides. That's not "nearly free"; that's a real line-item, and for a PoC the budget question is exactly the right one to ask.

Verdict: demote it. Core confound handling stays as it is in your doc — last-token measurement plus the transfer test, with transfer carrying the weight. Put the positional profile in Next Steps with one honest line: "positional analysis of where in the prompt the identity signal arises and whether it persists — descriptive of the model's in-context inference dynamics, but not required for H1/H2, and complicated by cross-position activation statistics."

And yes — the token-colored generations are the right visual, kept exactly as scoped last message: conditional on results, qualitative, high communicative value, near-zero methodological



risk. It has none of the positional profile's problems because it's not making an inferential claim — it's ILLUSTRATING a claim your quantitative results have already established, the same role the figure plays in Kim et al. (who explicitly flag theirs as qualitative). One prompt, one forward pass over the generation, color by projection, done. Put it in planned figures as "Figure 6 (conditional on H2b): scaffold-direction token coloring of Gemma generations, following Kim et al. Fig. 1."

So the clean division: transfer test = the argument; token coloring = the picture; positional profile = future work. That's also the right shape for a PoC generally — every analysis should either decide a hypothesis or communicate a decided one, and anything that does neither goes in Next Steps no matter how cheap it looks.

## User

can you explain kim et al clearly

## Claude

Here's the paper start to finish, in the order the argument builds.

The question. LLMs can convincingly write from liberal or conservative perspectives when asked. Kim, Evans & Schein ask HOW: is there a general, linear representation of the liberal-conservative axis inside the model's activations — a single direction in vector space such that "more conservative" means "further along that direction"? And if it exists, can you use it to monitor and control the political slant of generated text?

The measurement trick. To probe for a concept you need ground-truth labels, and "political perspective" is fuzzy. Their solution imports a validated instrument from political science: DW-NOMINATE, which scores every US lawmaker on a liberal-conservative axis from -1 to +1 based purely on their roll-call voting record. It's the standard ideology measure in the field, decades old, well validated. This converts a vague concept into 552 precise numbers — one per member of the 116th Congress (chosen deliberately because 2019-2021 predates Llama-2's training cutoff, so the models have seen these people).

The setup. For each lawmaker, they build one prompt: "USER: Generate a statement by [NAME], a politician in the United States. ASSISTANT: In 2019, [NAME] said that" — and run it through three open 7B chat models (Llama-2-7b-chat, Mistral-7b-instruct, Vicuna-7b). At the last token, they record the output of EVERY attention head separately: 32 layers x 32 heads = 1,024 activation vectors per prompt. Then, for each head independently, they fit a ridge regression ( $\lambda=1$ , chosen by cross-validation) predicting the lawmaker's DW-NOMINATE score from that head's activation, and evaluate by Spearman rank correlation on held-out lawmakers (2-fold CV).

Result 1: the representation exists and lives in the middle layers. Many heads in layers ~10-20 predict held-out lawmakers' ideology with Spearman  $\sim 0.8$ ; the best single heads hit 0.846-0.861 depending on the model. Ensembling the top 32 heads pushes this to  $\sim 0.87$ -0.89. Early and late layers are much weaker — the concept is COMPUTED at mid-depth, not present from the start. (That's their Figure 2, the layer-by-head heatmaps.)

Result 2: it's genuinely linear. Two robustness checks. First, they refit with one-hidden-layer MLP probes — nonlinear — and get essentially no improvement (e.g., 0.854 linear vs. 0.855 nonlinear for Llama), which is the standard evidence for "the concept is a direction, not a curved manifold." Second, a clever sanity check on whether probes can fit anything: they retrain probes on transformed labels. Randomly permuted scores  $\rightarrow$  correlation  $\sim 0.15$  (chance). A non-monotonic transform,  $\sin(10y) \rightarrow$  probes fail badly ( $\sim 0.5$ ). So the activations don't contain arbitrary



information — they specifically contain the ideological ordering.

Result 3 — the important one: the direction generalizes beyond its training data. This is the paper's real contribution, because the obvious objection to Result 1 is memorization: the prompts contain lawmaker NAMES, and DW-NOMINATE scores are on the internet, so maybe the probe just reads off a memorized lookup. Two answers. (a) Token-by-token monitoring: they have models write open-ended essays on policy issues — no names anywhere — and apply the trained probe to each generated token's activations. The readings are interpretable: the probe swings liberal on "those who support immigration argue it can bring many benefits" and conservative on "abortion is the intentional termination of a human life" (their Figure 1, the blue/red one you're reproducing). (b) Transfer to a different domain with different ground truth: prompting "[OUTLET] reported that" for 400 news outlets and applying the SAME probes — trained only on lawmakers, never retrained — predicts the outlets' Ad Fontes media-bias scores with Spearman 0.72-0.80. A name-score lookup table can't do that. The probes found the axis itself.

Result 4: the direction is causal. Following the inference-time intervention recipe (Li et al. 2024), they use the probe weight vectors as steering vectors: during generation, add  $\alpha$  times  $\hat{\sigma}$  times  $\hat{\theta}$  into the top-K predictive heads at every token ( $\hat{\sigma}$  normalizes by activation scale;  $\alpha$  controls magnitude and sign). They generate 1,134 essays across nine policy issues, sweeping  $\alpha$  in  $\{-30, \dots, +30\}$  and K in  $\{16, \dots, 96\}$ , and score each essay's slant on a 1-7 scale using GPT-4o — validated against ten politically balanced human annotators (3 D, 4 I, 3 R) with an intraclass correlation of 0.91. Findings: slant tracks  $\alpha$  monotonically for all three models (correlation up to 0.607 for Llama); at  $\alpha=0$  all three models sit slightly liberal of the scale's midpoint;  $|\alpha|>30$  breaks coherence; more heads (larger K) steers more effectively; steering works best on issues with a wide stance spectrum (immigration, abortion) and weakest on things like "trust in science"; interventions at early-to-middle layers work while middle-to-late layers do essentially nothing; and it keeps working on post-cutoff topics (the 2023 UAW strike), another anti-memorization point.

Limitations they own: 7B models only; US-centric single left-right axis (they note less polarized political systems might need a simplex of archetypes rather than one line); no claim their extraction method is optimal; and the GPT-4o judge, while validated, is still an LLM rating politics.

The one-paragraph compression: LLMs contain a linear liberal-conservative direction, computed in middle layers, discoverable by ridge-probing attention heads against lawmakers' DW-NOMINATE scores. The direction is not memorization — it transfers zero-shot to predicting news-outlet slant and reads sensibly off open-ended generation token-by-token — and it is causal: adding it to activations steers generated text's slant monotonically with dose.

Map to your project, piece by piece: their probing recipe is your probing recipe minus the head-level resolution (you probe the summed residual stream — sufficient, coarser, 26 sites instead of 1,024, no top-K selection machinery). Their existence + linearity + mid-layer localization results are the priors your H1 stands on. Their news-outlet transfer is the exact template for your token-detector defense — theirs proves the direction isn't a NAME detector, yours must prove it isn't a PREFIX detector, same inferential move. Their steering protocol ( $\alpha$  sweep, dose-response, GPT-4o judge with the ICC 0.91 validation) is your H2b protocol, minus the controls you're adding (random direction, specificity prompts) and with your own  $\alpha$  calibration since their  $\{-30 \dots +30\}$  window is Llama-specific. And the one thing they DON'T do is the thing you do: every direction in their paper is extracted from content or persona-as-author (lawmaker statements, outlet statements) — nobody asks whether telling the model who the USER is moves activations along that same axis. That sentence is your gap statement, and it's accurate.

## User

wait can you explain the predation of training cutoff

## Claude

Sure — "predates the training cutoff" is doing quiet but important work in two opposite directions in the paper, and it's easy to blur them.

The cutoff itself. A model's training data ends at some date — for Llama-2, roughly September 2022. The model has "read about" everything before that date and knows literally nothing after it (except what's in the prompt). This creates a clean natural experiment boundary: pre-cutoff facts are things the model COULD have memorized; post-cutoff facts are things it COULDN'T have.

Direction 1: the probing data must be BEFORE the cutoff. They chose the 116th Congress (2019-2021) deliberately so that every lawmaker they probe with is someone the model has extensively seen — news coverage, speeches, Wikipedia, all of it. Why is that required? Because the probing setup only works if the model has SOME internal representation of each lawmaker to read out. The prompt is just "Generate a statement by [NAME]" — all the political information has to come from what the model already associates with that name. If they'd used lawmakers elected in 2023, the model would have no idea who these people are, the activations would contain no ideology signal about them, and probe failure would be uninformative — you couldn't distinguish "no linear ideology representation exists" from "the model has never heard of these people." So pre-cutoff data isn't a convenience; it's a validity condition: it guarantees the ingredient (knowledge of the person) whose ORGANIZATION (linear ideology axis) is the thing under test.

The tension this creates. But the same fact that makes probing possible creates the memorization objection: DW-NOMINATE scores themselves are on the internet, pre-cutoff, so maybe the model rote-memorized "Senator X <-> score 0.43" and the probe is reading a lookup table rather than a general ideology axis. Pre-cutoff data is simultaneously what makes the experiment runnable and what makes the skeptical explanation available. That's why the paper needs the transfer results (news outlets, token-by-token essays) — those show the direction generalizes to targets and text the probe was never trained on.

Direction 2: the steering test is AFTER the cutoff — and that's the point. In their robustness checks they steer models to write about the 2023 UAW strike and the ADVANCE Act, both post-cutoff events. Here the logic flips: because the model has NO stored opinions, coverage, or partisan framings of these specific events, any liberal-vs-conservative slant in the steered output can't be retrieval of memorized takes about them — it has to be the model APPLYING its general ideological machinery to novel subject matter, pushed by the direction. Post-cutoff topics are the one setting where "it's just regurgitating memorized partisan content about this topic" is impossible by construction. So steering still working there is another independent strike against the memorization story, from the opposite side of the boundary.

The elegant summary: they use the cutoff as a scalpel twice — pre-cutoff data to GUARANTEE knowledge exists for the probe to read, post-cutoff topics to GUARANTEE knowledge doesn't exist so steering effects must be generalization. Same boundary, opposite exploitations, jointly closing the memorization objection from both ends.

Relevance to your PoC, briefly. Your setup mostly sidesteps direction 1 — "I am a Democrat" requires the model to know what a Democrat is, which is deep pre-cutoff general knowledge, not facts about specific individuals — but the Cen prompts do mention Biden/Harris/Trump 2024 content, and Gemma-2's cutoff means its knowledge of the 2024 race specifically is partial

(Gemma-2 was released mid-2024). That's worth one line of awareness in your doc: your matched pairs' SCAFFOLDS rely only on timeless category knowledge (safe), while some BASE QUESTIONS concern an election the model saw incompletely — fine for your purposes since the scaffold contrast holds the question constant, but a reader may ask. And if you ever want your own version of their direction-2 trick: steer Gemma on a political topic that postdates its cutoff and check that slant still responds — same anti-memorization dividend, one extra prompt set.

## User

*[The user attached the working document, v5 — with the corrected figure lead-in, inert-scaffold sentence, full Causal Eval controls, and consistent hyphenation.]*

how does this look:

## Claude

Nearly frozen. The figure lead-in, inert-scaffold sentence, and hyphenation all landed. What's still open is mostly items from last round that didn't make it in — flagging them once more briefly, since at this point they're the difference between "done" and "done":

1. Hypothesis nesting — still flat. "This PoC has two hypotheses" followed by four unlabeled lines still reads as four hypotheses. This is the one remaining substantive issue: label them H1 / H2 (umbrella) / H2a / H2b with indentation, and trim H1's "along interpretable directions" to just "measurable and systematic displacement of activations" — interpretability is H2a's and the SAE analysis's job, and H1 should be killable by a magnitude test alone.

2. Stubs still empty. Results = planned-figures list (probe-vs-layer curve; projection ranking vs. controls/nulls; partisanship correlation; SAE loading; dose-response + random control; conditional token-coloring figure). Next Steps = deferred items with reasons (positional profile, head attribution, 9B confirmation, token-level SAE, powered partisanship correlation, internal-vs-behavioral correlation). Five minutes of typing, and it's what makes the doc read as pre-registered.

3. Small ones, verbatim fixes: "Cen et al.." double period (intro, twice — end of first paragraph and in H2's umbrella line); H2a bullet 1 "i.e." -> "e.g."; Causal Eval "neutral political prompts" -> "open-ended political prompts" (consistency with your own spectrum stance); and add the one-line alpha caveat — Kim et al.'s {-30...+30} is calibrated to their models, so you'll locate Gemma's coherence cliff empirically rather than assuming their window.

4. Optional but cheap: the paraphrase check ("I am a Democrat" vs. "I'm a member of the Democratic party" should displace near-identically) as a third bullet in the token-confound list. One sentence, closes the surface-form objection preemptively.

Nothing else. The methodology chain is now complete and internally consistent: every hypothesis has a measurement, every measurement has a null, every causal claim has a control, and the deferred/conditional items are (or will be, per #2) explicitly parked rather than silently missing. Once the nesting and stubs are in, freeze it and stop editing — the next unit of progress is code, not prose.

On that: want the runnable skeleton? Concretely — HF dataset load, category filtering, base\_q\_template\_hash grouping into matched (scaffold, baseline) pairs, then the TransformerLens caching loop and the 26-probe sweep with the by-base-question CV split. That gets you to your first real figure (probe accuracy vs. layer), which is also the first place the project can surprise you.

**User**

can you turn this conversation into a pdf verbatim

**Claude**

*[Claude produced this PDF.]*